

Subsymbolic AI

Ibrahim Nasser*

Last updated on May 1, 2026

Contents

1	Introduction	1
2	Probability Theory	3
3	Bayesian Networks	15

Appendices

A	Mathematical Vernacular and Induction	21
A.1	Mathematical Language	21
A.2	Unary Natural Numbers	26
B	Set Theory	28
B.1	Sets	28
B.2	Relations	30
B.3	Functions	34
B.4	Equivalence Classes	40
C	Graphs and Trees	42
D	Mathematical Structures	45
E	Formal Languages and Grammars	47

*This document and its source files are licensed under Creative Commons Attribution–NonCommercial–NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

1 Introduction

Def 1.1. Artificial intelligence (AI): The capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making

Def 1.2. Symbolic AI: A subfield of [Artificial Intelligence \(AI\)](#) based on the assumption that many aspects of intelligence can be achieved by the manipulation of symbols, combining them into meaningful structures (expressions) and manipulating them (using processes) to produce new expressions.

Def 1.3. Statistical AI: Remedies the two shortcomings of [symbolic AI](#) approaches: that all concepts represented by symbols are crisply defined, and that all aspects of the world are knowable/representable in principle. Statistical AI adopts sophisticated mathematical models of uncertainty and uses them to create more accurate world models and reason about them.

Def 1.4. Subsymbolic AI (a.k.a connectionism/neural AI): A subfield of [AI](#) that posits that intelligence is inherently tied to brains, where information is represented by a simple sequence pulses that are processed in parallel via simple calculations realized by neurons, and thus concentrates on neural computing.

Def 1.5. Embodied AI: Posits that intelligence cannot be achieved by [reasoning](#) about the state of the world ([symbolically](#), [statistically](#), or [sub-symbolically](#)), but must be embodied i.e. situated in the world, equipped with a "body" that can interact with it via sensors and actuators. Here, the main method for realizing intelligent behavior is by learning from the world.

Def 1.6. Reasoning: The process of producing valid arguments and predictive world models. There are three forms of reasoning:

- **deductive reasoning** to produce new knowledge from existing knowledge. *Example:* All humans are mortal. Socrates is a human. Therefore, Socrates is mortal.
- **inductive reasoning** to produce knowledge from perception. *Example:* The sun has risen every morning in recorded history. Therefore, the sun will rise tomorrow.
- **abductive reasoning** to produce explanations for observations and given knowledge. *Example:* The grass is wet this morning. If it rained last night, that would explain it. Therefore, it probably rained.

Def 1.7. Inference: The act or process of reaching a **conclusion** about something from known facts or evidence (jointly called **premises**)

Def 1.8. Formal Logic: The science of [deductively](#) valid inferences, meaning arguments whose [conclusions](#) necessarily follow from their [premises](#) by virtue of their form (their structure), regardless of the topic.

Def 1.9. Agent: An agent is a [structure](#) $A := \langle \mathcal{P}, \mathcal{A}, f \rangle$ where:

- $\mathcal{P}$ is a [set](#) of percepts
- $\mathcal{A}$ is a [set](#) of actions
- f is a [function](#) $f : \mathcal{P}^* \rightarrow \mathcal{A}$ that maps from percepts to actions.

In other words, an agent is anything that perceives its environment via sensors and acts on it with actuators.

Def 1.10. Performance Measure: A [function](#) that evaluates a sequence of environments.

Def 1.11. Rational: An [agent](#) is called **rational**, if it chooses whichever action maximizes the expected value of the [performance measure](#) given the percept sequence to date.

Def 1.12. PEAS: To design [rational agents](#), we must specify the PEAS components: [Performance Measure](#), Environment, Actuators, and Sensors.

Def 1.13. Environment Types: For an [agent](#) a we classify the environment e of a by its **type**, which is one of the following. We call e

1. *fully observable*, iff the a 's sensors give it access to the complete state of the environment at any point in time; otherwise, we call it *partially observable*.
2. *deterministic*, iff the next state of the environment is completely determined by the current state and a 's actions; otherwise, *stochastic*.
3. *episodic*, iff a 's experience is divided into atomic episodes, where it perceives and then performs a single action, and the next episode does not depend on previous ones. Otherwise, *sequential*.
4. *dynamic*, iff the environment can change without an action performed by a ; otherwise, *static*. If the environment does not change but a 's performance measure does, we call e *semidynamic*.

5. *discrete*, iff the **sets** of e 's state and a 's actions are countable; otherwise, *continuous*.

6. *single-agent*, iff only a acts on e ; otherwise *multi-agent*.

Def 1.14. Reflex: An **agent** $\langle \mathcal{P}, \mathcal{A}, f \rangle$ is called a **reflex agent**, iff it only takes the last percept into account when choosing an action, i.e. $f(p_1, \dots, p_k) = f(p_k) \forall p_1 \dots p_k \in \mathcal{P}$

Def 1.15. Model-Based Agent: A model-based agent $\langle \mathcal{P}, \mathcal{A}, \mathcal{S}, \mathcal{T}, s_0, S, a \rangle$ is an **agent** $\langle \mathcal{P}, \mathcal{A}, f \rangle$ whose actions depend on:

- a **world model**: a **set** $\mathcal{S}$ of possible *states*, and a start state $s_0 \in \mathcal{S}$.
- a **transition model** $\mathcal{T}$ that predicts a new state $\mathcal{T}(s, a)$ from a state s and an action a .
- a **sensor model** S that given a state s and a percept p determines a new state $S(s, p)$.
- an **action function** $a : \mathcal{S} \rightarrow \mathcal{A}$ that given a state $s \in \mathcal{S}$ selects the next action $a \in \mathcal{A}$.

Note

If the agent is in state s then it took action a and now perceives p , then the agent state will become $s' = S(p, \mathcal{T}(s, a))$ and accordingly take action $a' = a(s')$.

Def 1.16. Goal Based Agent: A goal based agent $\langle \mathcal{P}, \mathcal{A}, \mathcal{S}, \mathcal{T}, s_0, S, a, \mathcal{G} \rangle$ is a **model based agent** with an explicit set of goals. It consists of:

- a set of internal states $\mathcal{S}$ and an initial state $s_0 \in \mathcal{S}$,
- a transition model $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$,
- a state update function $S : \mathcal{P} \times \mathcal{S} \rightarrow \mathcal{S}$,
- a set of goals $\mathcal{G}$,
- a goal conditioned action function $a : \mathcal{S} \times \mathcal{G} \rightarrow \mathcal{A}$ selecting an action given a state and the goals.

Def 1.17. Utility-Based Agent: A utility-based agent uses a world model along with a utility function that models its preferences among the states of that world. It chooses the action that maximizes the expected utility.

Def 1.18. Learning Agent: A learning agent is an **agent** that can improve its own behavior through experience. It is composed of four main components:

- the performance element, which selects actions based on the agent's current percepts and represents its existing knowledge or behavior,
- the learning element, which improves the performance element over time by analyzing feedback and identifying how to make better decisions,
- the critic, which evaluates the agent's performance according to a given performance standard and provides feedback to the learning element,
- the problem generator, which suggests new and informative actions or experiences that help the agent explore and learn more effectively.

Def 1.19. State Representation: We call a state representation **atomic**, iff it has no internal structure (black box). However, iff each state is characterized by attributes and their values then the representation is **factored**. A **structured** state representation is when include representations of objects, their properties and relationships.

2 Probability Theory

Def 2.1. Sample Space: We say that a **set** Ω is a **sample space** of a random experiment, iff Ω contains all possible outcomes of that experiment. Each individual outcome $\omega \in \Omega$ is called a **sample point** or an **elementary outcome**.

Example 2.1. Throwing two dice $\rightsquigarrow \Omega = \{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \dots, \langle 6, 6 \rangle\}$ with $\Gamma(\Omega) = 36$

Example 2.2. Number of tosses until the first head $\rightsquigarrow \Omega = \{1, 2, 3, \dots\} = \mathbb{N}$ with $\Gamma(\Omega) = \aleph_0$

Example 2.3. A random point in the interval $[0, 1] \rightsquigarrow \Omega = [0, 1]$ with $\Gamma(\Omega) = \mathfrak{c}$

Def 2.2. Event: Given a sample space Ω , we say that E is an **event** iff $E \subseteq \Omega$. That is, an event is a **subset** of the sample space. We say an event E occurs iff the outcome ω of the experiment satisfies $\omega \in E$.

Note

Because **events** are **sets**, we use Set Theory to describe relations between them:

- **Complement** $\bar{E}$: the **event** E that does not occur
- **Union** $A \cup B$: the **event** that A or B or both occurs
- **Intersection** $A \cap B$: the **event** that both A and B occur
- **Disjoint** / Mutually Exclusive: two **events** A and B are **disjoint** iff $A \cap B = \emptyset$

Note

We usually borrow propositional logic syntax to express **union** and **intersection**:

$$P(A \cup B) \rightsquigarrow P(A \vee B) \quad P(A \cap B) \rightsquigarrow P(A \wedge B)$$

We also use “,” to express **joint probability**

$$P(A \cap B) \rightsquigarrow P(A \wedge B) \rightsquigarrow P(A, B)$$

Example 2.4. The **sample space** for rolling one die is $\Omega = \{1, 2, \dots, 6\}$, **table 1** shows some **events** of interest.

Event Description	Set Representation
A : Even Number	$\{2, 4, 6\}$
B : Prime Number	$\{2, 3, 5\}$
$A \cap B$ (Even and Prime)	$\{2\}$
$A \cup B$ (Even or Prime)	$\{2, 3, 4, 5, 6\}$
$\bar{A}$ (Not Even)	$\{1, 3, 5\}$

Table 1: Examples of Events

Def 2.3. σ -algebra: Given a **sample space** Ω , a collection of **subsets** $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is called a **σ -algebra** iff it satisfies the following three axioms:

1. **Non-emptiness:** The sample space is included: $\Omega \in \mathcal{F}$. (which implies $\emptyset \in \mathcal{F}$ via complementation)
2. **Closure under complementation:** If $A \in \mathcal{F}$, then $\bar{A} \in \mathcal{F}$.
3. **Closure under countable unions:** If $A_1, A_2, \dots \in \mathcal{F}$ is a sequence of events, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**.

Example 2.5. Let the **sample space** be $\Omega = \{1, 2, 3\}$.

1. Consider the collection $\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$. This is a **σ -algebra** because:

- $\Omega \in \mathcal{F}$ (and $\emptyset \in \mathcal{F}$).
- It is closed under complements: $\{\bar{1}\} = \{2, 3\} \in \mathcal{F}$ and $\bar{\emptyset} = \Omega \in \mathcal{F}$.
- It is closed under unions: $\{1\} \cup \{2, 3\} = \Omega \in \mathcal{F}$.

2. Consider the collection $\mathcal{G} = \{\emptyset, \Omega, \{1\}, \{2\}\}$. This fails to be a σ -algebra for two reasons:

- No closure under Complements: $\{1\} \in \mathcal{G}$, but its complement $\{\bar{1}\} = \{2, 3\}$ is not in $\mathcal{G}$.
- No closure under Unions: $\{1\} \in \mathcal{G}$ and $\{2\} \in \mathcal{G}$, but their union $\{1, 2\}$ is not in $\mathcal{G}$.

Def 2.4. Probability Measure: A **probability measure** P is a function $P : \mathcal{F} \rightarrow [0, 1]$ that assigns a real number to every **event** $A \in \mathcal{F}$, satisfying the following **Kolmogorov Axioms**:

1. **Non-negativity:** $P(A) \geq 0$ for all $A \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable Additivity:** For any sequence of **disjoint events** $A_1, A_2, \dots \in \mathcal{F}$:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Def 2.5. Probability Space: A **probability space** is the triplet $\langle \Omega, \mathcal{F}, P \rangle$, where:

- Ω is the **sample space**.
- $\mathcal{F}$ is a σ -algebra on Ω .
- P is a **probability measure** defined on $(\Omega, \mathcal{F})$.

Example 2.6. For a fair 6-sided die roll, the **probability space** is defined as follows:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\mathcal{F} = \mathcal{P}(\Omega)$ (the power set)
- $P(A) = \frac{\Gamma(A)}{\Gamma(\Omega)} = \frac{\Gamma(A)}{6}$

Note

From the **Kolmogorov Axioms**, we can derive the following fundamental properties of **probability measures**:

- **Complement Rule:** For any **event** A , $P(\bar{A}) = 1 - P(A)$.
- **Monotonicity:** If $A \subseteq B$, then $P(A) \leq P(B)$.
- **Inclusion-Exclusion Principle:** For any two **events** A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Set Partition Identity:** For any two **events** A and B , B can be split into parts that are in A and parts that are not:

$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

- **Impossible Event:** $P(\emptyset) = 0$.

Def 2.6. Random Variable: Given a **probability space** $\langle \Omega, \mathcal{F}, P \rangle$, a **random variable** X is a **function** $X : \Omega \rightarrow D$ that maps each **sample point** $\omega \in \Omega$ to a numerical value in a **set** $D \subseteq \mathbb{R}$. We call D the **domain** of X , denoted by $\text{dom}(X)$.

Example 2.7. In a coin flip experiment where $\Omega = \{\text{Head}, \text{Tail}\}$, we can define a **random variable** X such that:

- $X(\text{Head}) = 1$
- $X(\text{Tail}) = 0$

In this case, $\text{dom}(X) = \{0, 1\}$.

Def 2.7. Probability of a Random Variable: For a **random variable** X and a specific value $x \in \text{dom}(X)$, the probability that X takes the value x is defined as the measure of the **event** consisting of all outcomes ω that map to x :

$$P(X = x) := P(\{\omega \in \Omega \mid X(\omega) = x\})$$

Example 2.8. Consider rolling a fair die where $\Omega = \{1, 2, \dots, 6\}$. If I define a **random variable** Y as the square of the outcome, then the probability of the **event** $Y = 4$ is:

$$P(Y = 4) = P(\{\omega \in \Omega \mid \omega^2 = 4\}) = P(\{2\}) = \frac{1}{6}$$

Def 2.8. Probability of Negation: For a **random variable** X and $x \in \text{dom}(X)$, we define the probability of the logical negation as:

$$P(X \neq x) := P(\{\omega \in \Omega \mid X(\omega) \neq x\}) = 1 - P(X = x)$$

Def 2.9. Probability of Disjunction: For **random variables** X_1, X_2 , we define the probability of the logical disjunction ("or") as:

$$P(X_1 = x_1 \vee X_2 = x_2) := P(\{\omega \in \Omega \mid X_1(\omega) = x_1\} \cup \{\omega \in \Omega \mid X_2(\omega) = x_2\})$$

Def 2.10. Joint Probability: Given two **random variables** X and Y defined on the same **sample space** Ω , the **joint probability** of $x \in \text{dom}(X)$ and $y \in \text{dom}(Y)$ is defined as the **probability** of the **intersection** of their individual **events**:

$$P(X = x, Y = y) := P(\{\omega \in \Omega \mid X(\omega) = x\} \cap \{\omega \in \Omega \mid Y(\omega) = y\})$$

Example 2.9. In a roll of two dice where X is the first die and Y is the second die: The joint probability $P(X = 3, Y = 4)$ is the probability of the **sample point** $(3, 4) \in \Omega$:

$$P(X = 3, Y = 4) := P(\{(3, 4)\}) := \frac{1}{36}$$

Example 2.10. To find the probability that the sum of two dice is 7 but neither is 3, we define the following **random variables** on the **sample space** $\Omega = \{1, \dots, 6\}^2$:

- X_1, X_2 : outcomes of the first and second die respectively.
- S : the sum, where $S(\omega) = X_1(\omega) + X_2(\omega)$.

We are interested in the **event** E :

$$E = \{\omega \in \Omega \mid S(\omega) = 7\} \cap \{\omega \in \Omega \mid X_1(\omega) \neq 3\} \cap \{\omega \in \Omega \mid X_2(\omega) \neq 3\}$$

Step 1: Identify outcomes where $S = 7$:

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

Step 2: Apply the constraints $X_1 \neq 3$ and $X_2 \neq 3$:

- $\omega \in A$ and $X_1(\omega) = 3 \implies \omega = (3, 4)$
- $\omega \in A$ and $X_2(\omega) = 3 \implies \omega = (4, 3)$

Step 3: Calculate the **cardinality** of the filtered set:

$$E = A \setminus \{(3, 4), (4, 3)\} = \{(1, 6), (2, 5), (5, 2), (6, 1)\} \implies \Gamma(E) = 4$$

Step 4: Use the **probability measure**:

$$P(E) = \frac{\Gamma(E)}{\Gamma(\Omega)} = \frac{4}{36} = \frac{1}{9}$$

Def 2.11. Marginalization: For two **random variables** X and Y , the **marginal probability** of X is calculated by summing the **joint probabilities** over all $y \in \text{dom}(Y)$:

$$P(X = x) := \sum_{y \in \text{dom}(Y)} P(X = x, Y = y)$$

This is a direct application of the axiom of countable additivity to the partition of the **event** $\{X = x\}$ into smaller, mutually exclusive joint events.

Example 2.11. Suppose I have a **joint probability** table for two **random variables** X and Y representing a simplified weather/activity model:

	$Y = 0$ (Stay)	$Y = 1$ (Walk)	Total
$X = 0$ (Rain)	0.3	0.1	$P(X = 0) = \mathbf{0.4}$
$X = 1$ (Sun)	0.1	0.5	$P(X = 1) = \mathbf{0.6}$

Table 2: Marginalization Example

To find $P(X = 0)$, I marginalize over Y :

$$P(X = 0) = P(X = 0, Y = 0) + P(X = 0, Y = 1) = 0.3 + 0.1 = 0.4$$

Def 2.12. Conditional Probability: Let A and B be **events** in a **probability space** where $P(B) > 0$. The **conditional probability** of A given B is defined as:

$$P(A | B) := \frac{P(A \cap B)}{P(B)}$$

In this context, we refer to:

- $P(A)$ as the **prior** probability.
- $P(B)$ as the **evidence**.
- $P(A | B)$ as the **posterior** probability.

Example 2.12. Suppose I roll a fair 6-sided die, so $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let A be the **event** that the outcome is 2, and B be the event that the outcome is even.

- $P(A) = P(\{2\}) = 1/6$
- $P(B) = P(\{2, 4, 6\}) = 3/6 = 1/2$
- $P(A \cap B) = P(\{2\}) = 1/6$

The probability that I roll a 2 **given** I know the result is even is:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/2} = \frac{1}{3}$$

Theorem 2.1. $P(A | C) = 1 - P(\bar{A} | C)$

Proof.

1. Partition: $C = (A \cap C) \cup (\bar{A} \cap C)$.
2. Note that $(A \cap C)$ and $(\bar{A} \cap C)$ are **disjoint** because $A \cap \bar{A} = \emptyset$.
3. Apply the **Axiom of Additivity**: $P(C) = P(A \cap C) + P(\bar{A} \cap C)$.
4. Rearrange: $P(\bar{A} \cap C) = P(C) - P(A \cap C)$.
5. **Conditional probability**:

$$P(\bar{A} | C) = \frac{P(\bar{A} \cap C)}{P(C)}$$

6. Substitute:

$$P(\bar{A} | C) = \frac{P(C) - P(A \cap C)}{P(C)}$$

7. Simplify:

$$P(\bar{A} | C) = \frac{P(C)}{P(C)} - \frac{P(A \cap C)}{P(C)} = 1 - P(A | C)$$

8. Rearrange the terms:

$$P(A | C) = 1 - P(\bar{A} | C)$$

□

Theorem 2.2. $P(A | C) = P(A \cap B | C) + P(A \cap \bar{B} | C)$

Proof.

1. Partition: $(A \cap C) = (A \cap C \cap B) \cup (A \cap C \cap \bar{B})$
2. Note that $(A \cap C \cap B)$ and $(A \cap C \cap \bar{B})$ are **disjoint** because $B \cap \bar{B} = \emptyset$.
3. **Axiom of Additivity:** $P(A \cap C) = P(A \cap B \cap C) + P(A \cap \bar{B} \cap C)$
4. **Conditional probability:**

$$P(A | C) = \frac{P(A \cap C)}{P(C)}$$

5. Substitute:

$$P(A | C) = \frac{P(A \cap B \cap C) + P(A \cap \bar{B} \cap C)}{P(C)}$$

6. Distribute:

$$P(A | C) = \frac{P(A \cap B \cap C)}{P(C)} + \frac{P(A \cap \bar{B} \cap C)}{P(C)}$$

7. Finally:

$$P(A | C) = P(A \cap B | C) + P(A \cap \bar{B} | C)$$

□

Theorem 2.3. $P(A \cup B | C) = P(A | C) + P(B | C) - P(A \cap B | C)$

Proof.

1. **Conditional probability**

$$P(A \cup B | C) = \frac{P((A \cup B) \cap C)}{P(C)}$$

2. Distribute: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
3. Inclusion-Exclusion:

$$P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - P((A \cap C) \cap (B \cap C))$$

4. Simplify: $(A \cap C) \cap (B \cap C) = A \cap B \cap C$
5. Substitute:

$$P(A \cup B | C) = \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)}$$

6. Simplify:

$$P(A \cup B | C) = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$$

7. Finally:

$$P(A \cup B | C) = P(A | C) + P(B | C) - P(A \cap B | C)$$

□

Theorem 2.4.

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

(Bayes' Theorem)

Proof.

1. **conditional probability:**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

2. Multiply both sides by $P(B)$ to express the **joint probability**:

$$P(A \cap B) = P(A | B) \cdot P(B)$$

3. **conditional probability**:

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

4. Multiply both sides by $P(A)$ to get a second expression for the **joint probability**:

$$P(B \cap A) = P(B | A) \cdot P(A)$$

5. **intersection** is commutative $\rightsquigarrow P(A \cap B) = P(B \cap A)$

6. Equate the right-hand sides of the expressions from Step 2 and Step 4:

$$P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

7. Divide both sides by $P(B)$ (assuming $P(B) > 0$) to isolate the **posterior** probability:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

□

Def 2.13. Bayesian Terminology: Within the context of **Bayes' Theorem**, we define the following nomenclature:

- **Posterior:** $P(A | B)$ is the probability of the hypothesis A after observing the evidence B .
- **Likelihood:** $P(B | A)$ is the probability that the evidence B would be observed if the hypothesis A were true.
- **Prior:** $P(A)$ is the probability of the hypothesis A before the evidence B is taken into account.
- **Evidence:** $P(B)$ is the total probability of observing the evidence under all possible hypotheses.

Def 2.14. The Product Rule: For any two **events** A and B where the **conditional probabilities** are well-defined ($P(A) > 0, P(B) > 0$), the probability of their **joint occurrence** is:

$$P(A \cap B) = P(A | B) \cdot P(B)$$

$$P(A \cap B) = P(B | A) \cdot P(A)$$

This rule establishes that the probability of "A and B" is the probability of one event, scaled by the probability of the other event occurring given that the first has occurred.

Example 2.13. Suppose I draw two cards from a deck without replacement. Let A be the event the first card is an Ace, and B be the event the second card is an Ace.

- $P(A) = 4/52$
- $P(B | A) = 3/51$ (since one Ace is gone)

The probability of drawing two Aces is:

$$P(A \cap B) = P(B | A) \cdot P(A) = \frac{3}{51} \cdot \frac{4}{52}$$

Def 2.15. Independence: Two **events** A and B are **independent** iff $P(A \cap B) = P(A) \cdot P(B)$

Theorem 2.5. If A and B are **independent**, then the **posterior** equals the **prior**:

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B)$$

Proof.

1. **Conditional probability:**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

2. **Independence:**

$$P(A | B) = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

3. The proof for $P(B | A) = P(B)$ follows identical logic by symmetry. □

Def 2.16. Pairwise Independence: A collection of **events** $A_1, A_2, \dots, A_n$ is **pairwise independent** iff every pair of distinct events is **independent**. That is, for all $i \neq j$:

$$P(A_i \cap A_j) = P(A_i) \cdot P(A_j)$$

Def 2.17. Mutual Independence: A collection of **events** $A_1, A_2, \dots, A_n$ is **mutually independent** iff for every finite subset of indices $I \subseteq \{1, 2, \dots, n\}$, the probability of their **intersection** is the product of their individual probabilities:

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Note

Mutual independence strictly implies **pairwise independence**, but **pairwise independence** does not guarantee **mutual independence**.

Def 2.18. Conditional Independence: Given events A, B, C with $P(C) \neq 0$, then A and B are **conditionally independent** given C iff $P(A | B \cap C) := P(A | C)$, or $P(B | A \cap C) := P(B | C)$:

Theorem 2.6. If A and B are **conditionally independent** given C , then:

$$P(A \cap B | C) := P(A | C) \cdot P(B | C)$$

Proof.

1. By **definition**:

$$P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)}$$

2. By the **product rule**:

$$P(A \cap B \cap C) = P(A | B \cap C) \cdot P(B \cap C)$$

3. Expanding $P(B \cap C)$ further:

$$P(A \cap B \cap C) = P(A | B \cap C) \cdot P(B | C) \cdot P(C)$$

4. Substituting Step 3 into Step 1:

$$P(A \cap B | C) = \frac{P(A | B \cap C) \cdot P(B | C) \cdot P(C)}{P(C)}$$

5. Simplifying:

$$P(A \cap B | C) = P(A | B \cap C) \cdot P(B | C)$$

6. Since A and B are **conditionally independent** given C , $P(A | B \cap C) = P(A | C)$, yielding:

$$P(A \cap B | C) = P(A | C) \cdot P(B | C)$$

□

Def 2.19. Probability Distribution: A **probability distribution** is a vector v with values in $[0, 1]$ such that $\sum_i v_i = 1$.

Note

The **probability distribution** of a **random variable** X (written as $\mathbf{P}(X)$) is a complete description of the **probabilities** $P(X = x)$ for all values $x \in \text{dom}(X)$. It completely characterizes the behavior of the random variable by mapping each possible outcome to its corresponding probability, effectively describing how the total probability of 1 (from the **Normalization axiom**) is distributed across the domain of X .

Def 2.20. Full Joint Probability Distribution (JPD): Given a set of **random variables** $X_1, X_2, \dots, X_n$, a **full joint probability distribution** $\mathbf{P}(X_1, X_2, \dots, X_n)$ is a **probability distribution** that assigns a probability to every possible, mutually exclusive **joint probability** assignment for all variables. It can be represented as an n -dimensional tensor where each entry $P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) \in [0, 1]$ corresponds to a single atomic event, and the sum over all possible combinations of values is 1:

$$\sum_{x_1 \in \text{dom}(X_1)} \cdots \sum_{x_n \in \text{dom}(X_n)} P(X_1 = x_1, \dots, X_n = x_n) = 1$$

Example 2.14. Consider two boolean **random variables**: W (Weather: Sunny=1, Rainy=0) and C (Commute: Walk=1, Drive=0). The **full joint probability distribution** is given by the table:

	$C = \text{Walk}$	$C = \text{Drive}$	Total
$W = \text{Sunny}$	0.6	0.1	0.7
$W = \text{Rainy}$	0.05	0.25	0.3
Total	0.65	0.35	1.0

Note that the sum of all internal cells ($0.6 + 0.1 + 0.05 + 0.25 = 1.0$) satisfies the normalization requirement.

Def 2.21. Conditional Probability Distribution (CPD): Given **random variables** X and Y , the **conditional probability distribution** $\mathbf{P}(X \mid Y)$ represents a complete table or matrix of **probabilities**, providing the distribution of X for every possible observed value $y \in \text{dom}(Y)$. It consists of the **conditional probabilities** $P(X = x \mid Y = y)$ for all $x \in \text{dom}(X)$ and all $y \in \text{dom}(Y)$. For any fixed y , the distribution over X satisfies the normalization constraint:

$$\sum_{x \in \text{dom}(X)} P(X = x \mid Y = y) = 1$$

Example 2.15. Continuing from the previous example, we can form the **conditional probability distribution** $\mathbf{P}(C \mid W)$ by dividing each joint probability by the marginal probability of W . Note that for any fixed value of W (each column), the probabilities sum to 1:

$\mathbf{P}(C \mid W)$	$W = \text{Sunny}$	$W = \text{Rainy}$
$C = \text{Walk}$	$0.6/0.7 = 6/7$	$0.05/0.3 = 1/6$
$C = \text{Drive}$	$0.1/0.7 = 1/7$	$0.25/0.3 = 5/6$
Total	1.0	1.0

Def 2.22. A **Naive Bayes Model** (Bayesian Classifier) consists of:

- **Random variables** $E_1 \cdots E_n$ (effects)
- **Random variable** C (the cause)
- The **probability distribution** $\mathbf{P}(C)$
- The **CPD** $\mathbf{P}(E_i \mid C)$

Where all $E_1, \dots, E_n$ are **conditionally independent** given C .

Lemma 2.7. Chain Rule:

$$\mathbf{P}(X_1, \dots, X_n) = \mathbf{P}(X_1 \mid X_2, \dots, X_n) \cdot \mathbf{P}(X_2 \mid X_3, \dots, X_n) \cdots \mathbf{P}(X_{n-1} \mid X_n) \cdot \mathbf{P}(X_n)$$

Proof. via generalization of the **product rule** for e.g. $X_1, \dots, X_4$:

- $\mathbf{P}(X_1, X_2, X_3, X_4) = \mathbf{P}(X_1 \mid X_2, X_3, X_4) \cdot \mathbf{P}(X_2, X_3, X_4)$
- $\mathbf{P}(X_2, X_3, X_4) = \mathbf{P}(X_2 \mid X_3, X_4) \cdot \mathbf{P}(X_3, X_4)$
- $\mathbf{P}(X_3, X_4) = \mathbf{P}(X_3 \mid X_4) \cdot \mathbf{P}(X_4)$
- Hence, $\mathbf{P}(X_1, X_2, X_3, X_4) = \mathbf{P}(X_1 \mid X_2, X_3, X_4) \cdot \mathbf{P}(X_2 \mid X_3, X_4) \cdot \mathbf{P}(X_3 \mid X_4) \cdot \mathbf{P}(X_4)$

□

Question

Given a **Naive Bayes Model**, can we compute the Full **JPD** $\mathbf{P}(C, E_1, \dots, E_n)$ using only $\mathbf{P}(C)$ and $\mathbf{P}(E_i \mid C)$?

Answer

Yes. Because of the **conditional independence** assumption in a **Naive Bayes Model**, the **JPD** simplifies to:

$$\mathbf{P}(C, E_1, \dots, E_n) = \mathbf{P}(C) \prod_{i=1}^n \mathbf{P}(E_i \mid C)$$

Proof.

- **Chain rule:** $\mathbf{P}(E_1, \dots, E_n, C) = \mathbf{P}(E_1 \mid E_2, \dots, E_n, C) \cdots \mathbf{P}(E_n \mid C) \cdot \mathbf{P}(C)$
- All $E_1, \dots, E_n$ are **conditionally independent** given C
- $\mathbf{P}(E_1 \mid E_2, \dots, E_n, C) = \mathbf{P}(E_1 \mid C)$ and $\mathbf{P}(E_2 \mid E_3, \dots, E_n, C) = \mathbf{P}(E_2 \mid C)$ and so on $\cdots$
- Hence, $\mathbf{P}(C, E_1, \dots, E_n) = \mathbf{P}(C) \prod_{i=1}^n \mathbf{P}(E_i \mid C)$

□

Def 2.23. Marginalization Generalized: Given **random variables** $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$, we have

$$\mathbf{P}(X_1, \dots, X_n) = \sum_{y_1 \in \text{dom}(Y_1), \dots, y_m \in \text{dom}(Y_m)} \mathbf{P}(X_1, \dots, X_n, Y_1 = y_1, \dots, Y_m = y_m)$$

Def 2.24. Given a **Naive Bayes Model** with a set of effects $E = \{E_1, \dots, E_n\}$, we refer to the **observable** effect as $O = \{O_1, \dots, O_{no}\}$ and the set of **unobservable** (unknown) effects as $U = \{U_1, \dots, U_{nu}\}$ where $E = O \cup U$

Notation Understanding

Assume we have two unknown effects ($nu = 2$): *Headache* (U_1) where $D_H = \{\text{yes}, \text{no}\}$ and *Fever* (U_2) where $D_F = \{\text{high}, \text{low}\}$

We define $D_u = \text{dom}(U_1) \times \dots \times \text{dom}(U_{nu}) = \{(y, h), (y, l), (n, h), (n, l)\}$

We can sum all [joint probabilities](#) of the unknown effects as :

$$\sum_{u \in D_u} P(U_1 = u_1, \dots, U_{nu} = u_{nu}) = 1$$

- For $u = (y, h)$, we have $P(U_1 = y, U_2 = h)$
- For $u = (y, l)$, we have $P(U_1 = y, U_2 = l)$
- For $u = (n, h)$, we have $P(U_1 = n, U_2 = h)$
- For $u = (n, l)$, we have $P(U_1 = n, U_2 = l)$

For a specific tuple $u = \langle u_1, u_2 \rangle \in D_u$, assuming *Headache* and *Fever* are [independent](#), the [joint probability](#) of $P(H = u_1, F = u_2)$ can be also computed from the product:

$$\prod_{j=1}^{nu} P(U_j = u_{u_j}) \quad \text{where } u_{u_j} \text{ means the value little } u \text{ at index } j \text{ given the tuple } u$$

Let's take the tuple $\langle n, h \rangle$ and we are iterating over $U = \{H(U_1), F(U_2)\}$

$$\prod_{j=1}^{nu} P(U_j = u_{u_j}) = P(H = n) \cdot P(F = h)$$

If we want to sum all the [joint probabilities](#) for all values, we write:

$$\sum_{u \in D_u} \prod_{j=1}^{nu} P(U_j = u_{u_j})$$

Which will evaluate to 1 too (assuming [independence](#)).

Let $D_u = \text{dom}(U_1) \times \dots \times \text{dom}(U_{nu})$, then:

$$\mathbf{P}(C \mid O_1, \dots, O_{no}) = \frac{\mathbf{P}(C, O_1, \dots, O_{no})}{\mathbf{P}(O_1, \dots, O_{no})}$$

We [marginalize](#) over the unknowns to compute the numerator:

$$\mathbf{P}(C, O_1, \dots, O_{no}) = \sum_{u \in D_u} \mathbf{P}(C, O_1, \dots, O_{no}, U_1 = u_1, \dots, U_{nu} = u_{nu})$$

$$\mathbf{P}(C, O_1, \dots, O_{no}) = \sum_{u \in D_u} \mathbf{P}(C) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C) \prod_{j=1}^{nu} \mathbf{P}(U_j = u_{u_j} \mid C)$$

$$\mathbf{P}(C, O_1, \dots, O_{no}) = \mathbf{P}(C) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C) \sum_{u \in D_u} \prod_{j=1}^{nu} \mathbf{P}(U_j = u_{u_j} \mid C)$$

Also, for the denominator, we have:

$$\mathbf{P}(O_1, \dots, O_{no}) = \sum_{c \in \text{dom}(C)} \sum_{u \in D_u} \mathbf{P}(O_1, \dots, O_{no}, C = c, U_1 = u_1, \dots, U_{nu} = u_{nu})$$

$$\mathbf{P}(O_1, \dots, O_{no}) = \sum_{c \in \text{dom}(C)} \sum_{u \in D_u} P(C = c) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C = c) \prod_{j=1}^{nu} P(U_j = u_{u_j} \mid C = c)$$

$$\mathbf{P}(O_1, \dots, O_{no}) = \sum_{c \in \text{dom}(C)} P(C = c) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C = c) \sum_{u \in D_u} \prod_{j=1}^{nu} P(U_j = u_{u_j} \mid C = c)$$

Therefore,

$$\mathbf{P}(C \mid O_1, \dots, O_{no}) = \frac{\mathbf{P}(C) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C) \sum_{u \in D_u} \prod_{j=1}^{nu} \mathbf{P}(U_j = u_{u_j} \mid C)}{\sum_{c \in \text{dom}(C)} P(C = c) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C = c) \sum_{u \in D_u} \prod_{j=1}^{nu} P(U_j = u_{u_j} \mid C = c)}$$

Note

$$\sum_{u \in D_u} \prod_{j=1}^{nu} P(U_j = u_{u_j} \mid C = c) = 1$$

Hence,

$$\mathbf{P}(C \mid O_1, \dots, O_{no}) = \frac{\mathbf{P}(C) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C)}{\sum_{c \in \text{dom}(C)} P(C = c) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C = c)}$$

Note

The denominator is

- the same for any given observations $O_1, \dots, O_{no}$ independent of the value of C
- the sum over all numerators

The denominator acts as a **normalizing constant**. It scales the product in the numerator so that the resulting values for $\mathbf{P}(C \mid O_1, \dots, O_{no})$ sum to 1. This process is called **Normalization**.

Def 2.25. Normalization: Given a vector $w := \langle w_1, \dots, w_k \rangle$ where each entry $w_i \in [0, 1]$ and $\sum_{i=1}^k w_i \leq 1$, we define a **function** $\alpha(w)$ that normalizes the vector by dividing each entry by the sum of all its components.

$$(\alpha(w))_i = \frac{w_i}{\sum_{j=1}^k w_j}$$

By construction, $\sum_{i=1}^k (\alpha(w))_i = 1$, which ensures that $\alpha(w)$ is a valid **probability distribution**.

Inference Theorem in a Naive Bayes Model

$$\mathbf{P}(C \mid O_1, \dots, O_{no}) = \alpha \left(\mathbf{P}(C) \prod_{i=1}^{no} \mathbf{P}(O_i \mid C) \right)$$

Example 2.16. Consider we are modeling *cavity* (C) and *toothache* (T) with Boolean domains $\{T, F\}$ and we know the following probabilities:

- $P(C = T) = 0.1$ $P(T = T \mid C = T) = 0.8$ $P(T = T \mid C = F) = 0.05$

What is the probability distribution for a patient having *cavity* given they have *toothache*?

$$\begin{aligned} \mathbf{P}(C \mid t = T) &:= \alpha(\mathbf{P}(C) \cdot \mathbf{P}(T = T \mid C)) \\ \mathbf{P}(C) &:= \langle P(C = T), P(C = F) \rangle \\ \mathbf{P}(T = T \mid C) &:= \langle P(T = T \mid C = T), P(T = T \mid C = F) \rangle \\ \mathbf{P}(C \mid t = T) &:= \alpha(\langle P(C = T), P(C = F) \rangle \cdot \langle P(T = T \mid C = T), P(T = T \mid C = F) \rangle) \\ &:= \alpha(\langle P(C = T) \cdot P(T = T \mid C = T), P(C = F) \cdot P(T = T \mid C = F) \rangle) \\ &:= \alpha(\langle 0.1 \cdot 0.8, 1 - P(C = T) \cdot 0.05 \rangle) \\ &:= \alpha(\langle 0.1 \cdot 0.8, 0.9 \cdot 0.05 \rangle) \\ &:= \alpha(\langle 0.08, 0.045 \rangle) \\ &:= \langle 0.64, 0.36 \rangle \end{aligned}$$

Theorem 2.8. Let $Q, E_1, \dots, E_{ne}, U_1, \dots, U_{nu}$ be **random variables** and $D_u = \text{dom}(U_1) \times \dots \times \text{dom}(U_{nu})$. Then:

$$\mathbf{P}(Q \mid E_1 = e_1, \dots, E_{ne} = e_{ne}) = \alpha \left(\sum_{u \in D_u} \mathbf{P}(Q, E_1 = e_1, \dots, E_{ne} = e_{ne}, U_1 = u_1, \dots, U_{nu} = u_{nu}) \right)$$

We call Q the **query variable**, $E_1, \dots, E_{ne}$ the **evidence**, and $U_1, \dots, U_{nu}$ the **unknown** (or **hidden**) **variables**, and computing a **conditional probability** this way **enumeration**

Note

General Law of Inference by Enumeration: "To find out what you care about (Q), you must look at every possible world (U) that could exist alongside what you already saw (E).” concretely:

- sum over all relevant entries of the **JPD** of the variables, and
- normalize the results to yield a **probability distribution**

3 Bayesian Networks

Def 3.1. A **Bayesian Network** consists of

1. a **DAG** $\langle \mathcal{X}, E \rangle$ of **random variables** $\mathcal{X} = \{X_1, \dots, X_n\}$, and
2. a **CPD** $\mathbf{P}(X_i \mid \text{Parents}(X_i))$ for every $X_i \in \mathcal{X}$,

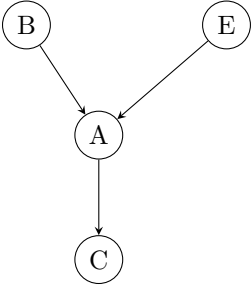
such that every X_i is **conditionally independent** of any conjunctions of **non-descendants** of X_i given $\text{Parents}(X_i)$

Note

Roots of the **DAG** are **conditionally independent** given the empty set $\emptyset$; i.e. they are **independent**. This follows directly from the definition:

- By definition, a root node X_i has no parents, meaning $\text{Parents}(X_i) = \emptyset$.
- Any other root node X_j is a **non-descendant** of X_i (since there are no directed paths between roots).
- By the definition of the **Bayesian Network**, X_i is conditionally independent of its non-descendant X_j given its parents ($\emptyset$).
- Conditional independence given the empty set is exactly equivalent to marginal **independence**.

Example 3.1. Consider a simple **Bayesian Network** modeling a security system with variables: B (Burglar), E (Earthquake), A (Alarm rings), and C (Call Police).



- Variables $\mathcal{X} = \{B, E, A, C\}$
- Edges $E = \{(B, A), (E, A), (A, C)\}$
- **CPDs**: $\mathbf{P}(B)$, $\mathbf{P}(E)$, $\mathbf{P}(A \mid B, E)$, and $\mathbf{P}(C \mid A)$

By definition, each node is **conditionally independent** of its **non-descendants** given its parents. Let's analyze each node:

- **Node B**: $\text{Parents}(B) = \emptyset$, $\text{NonDesc}(B) = \{E\}$.
Property: B is independent of E given $\emptyset \implies \mathbf{P}(B \mid E) = \mathbf{P}(B)$.
- **Node E**: $\text{Parents}(E) = \emptyset$, $\text{NonDesc}(E) = \{B\}$.
Property: E is independent of B given $\emptyset \implies \mathbf{P}(E \mid B) = \mathbf{P}(E)$.
- **Node A**: $\text{Parents}(A) = \{B, E\}$, $\text{NonDesc}(A) = \emptyset$.
Property: Vacuously true (no non-descendants exists to be independent from).
- **Node C**: $\text{Parents}(C) = \{A\}$, $\text{NonDesc}(C) = \{B, E\}$.
Property: C is independent of $\{B, E\}$ given $A \implies \mathbf{P}(C \mid A, B, E) = \mathbf{P}(C \mid A)$.
(Intuition: Once we know the alarm rang, knowing exactly **why** it rang doesn't give us any extra information about whether the police will be called).

Theorem 3.1. The **JPD** of a **Bayesian Network** $\langle \mathcal{X}, E \rangle$ is given by:

$$\mathbf{P}(X_1, \dots, X_n) = \prod_{X_i \in \mathcal{X}} \mathbf{P}(X_i \mid \text{Parents}(X_i))$$

Proof.

1. Let $X_1, X_2, \dots, X_n$ be a **topological sort** (an ordering) of the variables in the **DAG**. This ordering guarantees that if $X_i \rightarrow X_j$ is an edge (meaning X_i is a parent of X_j), then $i < j$.
2. Because joint probability is commutative, we can examine the variables in the **JPD** as $\mathbf{P}(X_n, \dots, X_1)$. Applying the **Chain Rule**, we get:

$$\begin{aligned} \mathbf{P}(X_n, \dots, X_1) &= \mathbf{P}(X_n \mid X_{n-1}, \dots, X_1) \cdot \mathbf{P}(X_{n-1} \mid X_{n-2}, \dots, X_1) \cdots \mathbf{P}(X_2 \mid X_1) \cdot \mathbf{P}(X_1) \\ &= \prod_{i=1}^n \mathbf{P}(X_i \mid X_{i-1}, \dots, X_1) \end{aligned}$$

3. Because we used a topological sort, all variables $X_{i-1}, \dots, X_1$ appear before X_i in the ordering. This guarantees they must be **non-descendants** of X_i (since descendants can only appear *after* X_i).
4. Thus, the conditioning set $\{X_{i-1}, \dots, X_1\}$ consists exactly of $\text{Parents}(X_i)$ plus some other non-descendants of X_i .
5. By the definition of **Bayesian Networks**, X_i is **conditionally independent** of its **non-descendants** given its parents. Therefore, the extra **non-descendants** drop out of the probability:

$$\mathbf{P}(X_i \mid X_{i-1}, \dots, X_1) = \mathbf{P}(X_i \mid \text{Parents}(X_i))$$

6. Substituting this back into the **Chain Rule** from Step 2 yields:

$$\mathbf{P}(X_n, \dots, X_1) = \prod_{i=1}^n \mathbf{P}(X_i \mid \text{Parents}(X_i))$$

By commutativity of the joint probability, this is equivalent to $\mathbf{P}(X_1, \dots, X_n)$. Validating the theorem. □

Def 3.2. Given **random variables** $X_1, \dots, X_n$ with **finite domains** $D_1, \dots, D_n$, the **size** of the **Bayesian Network** $\mathcal{B} = \langle \{X_1, \dots, X_n\}, E \rangle$ is defined as:

$$\text{size}(\mathcal{B}) := \sum_{i=1}^n |D_i| \cdot \left(\prod_{X_j \in \text{Parents}(X_i)} |D_j| \right)$$

Note

- $\text{size}(\mathcal{B}) \triangleq$ The total number of entries in the **CPD**.
- Smaller **Bayesian Networks** need to estimate less probabilities $\rightsquigarrow$ more efficient inference.
- Explicit Full **JPD** has size $\prod_{i=1}^n |D_i|$
- If $|\text{Parents}(X_i)| \leq k$ for every X_i , and $D_{\max}$ is the largest **random variable domain**, then $\text{size}(\mathcal{B}) \leq n |D_{\max}|^{k+1}$
- In the worst case, $\text{size}(\mathcal{B}) = \sum_{i=1}^n \prod_{j=1}^i |D_j|$, namely if every variable depends on all its predecessors in the chosen variable ordering.
- **Bayesian Networks** are compact if each variable is directly influenced only by few of its predecessor variables.

Example 3.2. Consider a **Bayesian Network** $\mathcal{B}$ with 4 boolean **random variables** (meaning their domain sizes are $|D_i| = 2$ for all i) arranged in a simple chain: $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$.

We execute the formula term by term:

- **Node** X_1 : No parents ($\text{Parents}(X_1) = \emptyset \implies$ independent product is 1).
Term expands to: $|D_1| \cdot 1 = 2 \cdot 1 = 2$
- **Node** X_2 : One parent ($\text{Parents}(X_2) = \{X_1\}$).
Term expands to: $|D_2| \cdot |D_1| = 2 \cdot 2 = 4$
- **Node** X_3 : One parent ($\text{Parents}(X_3) = \{X_2\}$).
Term expands to: $|D_3| \cdot |D_2| = 2 \cdot 2 = 4$
- **Node** X_4 : One parent ($\text{Parents}(X_4) = \{X_3\}$).
Term expands to: $|D_4| \cdot |D_3| = 2 \cdot 2 = 4$

Summing these up gives the total size of the probability tables to be stored:

$$\text{size}(\mathcal{B}) = 2 + 4 + 4 + 4 = 14 \text{ probability entries}$$

We can verify the efficiency by comparing it to an explicit Full **JPD**:

$$\text{Full JPD Size} = \prod_{i=1}^4 |D_i| = 2 \cdot 2 \cdot 2 \cdot 2 = 16 \text{ entries}$$

Even on just 4 variables, the conditional independencies in the **Bayesian Network** make it tangibly more compact.

Upper Bound Analysis

In our chain, no node has more than 1 parent, so the maximum in-degree is $k = 1$. The largest domain is $D_{\max} = 2$ (because all variables are boolean). For our $n = 4$ nodes, the sparsity upper bound formula yields:

$$\text{size}(\mathcal{B}) \leq n \cdot |D_{\max}|^{k+1} = 4 \cdot 2^{1+1} = 4 \cdot 4 = 16$$

Our actual calculated size of 14 is indeed strictly bounded by 16.

To see the real power of this bound, imagine extending this exact chain to $n = 100$ boolean variables. The Full JPD would require 2^{100} entries (a computational impossibility). In contrast, our [Bayesian Network](#) upper bound guarantees the memory footprint is simply $\leq 100 \cdot 2^2 = 400$ entries. This demonstrates why bounding the maximum number of parents k is critical for scalability!

Question

How to keep our [Bayesian Network](#) small?

Answer

- Reduce the number of edges $\rightsquigarrow$ Order the variables to allow for exploiting [conditional independence](#) (causes before effects), or
- Represent the [CPD](#) efficiently:
 - For a Boolean [random variable](#) X , we only need to store $P(X = T \mid \text{Parents}(X))$
 - Introduce different kinds of nodes exploiting domain knowledge.

Def 3.3. BN Construction Algorithm:

1. Initialize $BN = \langle \{X_1, \dots, X_n\}, E \rangle$ where $E = \emptyset$
2. Fix any variable ordering, $X_1, \dots, X_n$
3. **for** $i := 1, \dots, n$ **do**:
 - (a) Choose a minimal set $\text{Parents}(X_i) \subseteq \{X_1, \dots, X_{i-1}\}$ such that:

$$\mathbf{P}(X_i \mid X_{i-1}, \dots, X_1) = \mathbf{P}(X_i \mid \text{Parents}(X_i))$$

- (b) For each $X_j \in \text{Parents}(X_i)$, insert $\langle X_j, X_i \rangle$ into E
- (c) Associate X_i with $\mathbf{P}(X_i \mid \text{Parents}(X_i))$

Note

The size of the resulting [Bayesian Network](#) depends on the chosen **variable ordering**. The size of a [Bayesian Network](#) is not a fixed property (it depends on the skill of the designer)

Rule of Thumb

Order *causes* before *effects*

Def 3.4. A node X in a [Bayesian Network](#) is called **deterministic**, if its value is completely determined by the values of $\text{Parents}(X)$

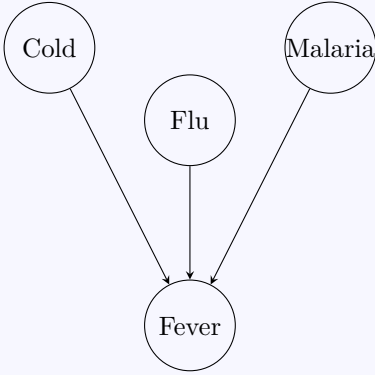
Example 3.3. The sum of two dice throws S is entirely determined by the values of the two dice **First** and **Second**

Note

- [Deterministic](#) nodes model direct, causal relationships.
- If X is [deterministic](#), then $P(X \mid \text{Parents}(X)) \in \{0, 1\}$
- We can replace the [CPD](#) $\mathbf{P}(X \mid \text{Parents}(X))$ by a boolean function

Noisy Nodes

Sometimes, values of nodes are *almost deterministic*. For example, assume we have a network that model all causes of fever:



If there is a fever, then one of them (at least) must be the cause, but none of them necessarily cause a fever. We say that the causal relation between parent and child is **inhibited**. We can model **inhibition** by individual inhibition factors q_d

Def 3.5. The **CPD** of a **noisy disjunction node** X with $\text{Parents}(X) = \{X_1, \dots, X_n\}$ in a **Bayesian Network** is given by:

$$P(X \mid X_1, \dots, X_n) = 1 - \left(\prod_{\{j \mid X_j = T\}} q_j \right)$$

where the q_i are the **inhibition factors** of $X_i \in \text{Parents}(X)$, defined as:

$$q_i := P(\neg X \mid \neg X_1, \dots, \neg X_{i-1}, X_i, \neg X_{i+1}, \dots, \neg X_n)$$

Note

Instead of a distribution with 2^k parameters, we only need k parameters.

Example 3.4. Assume the following **inhibition factors**:

$$\begin{aligned} q_{\text{cold}} &= P(\neg \text{fever} \mid \text{cold}, \neg \text{flu}, \neg \text{malaria}) = 0.6 \\ q_{\text{flu}} &= P(\neg \text{fever} \mid \neg \text{cold}, \text{flu}, \neg \text{malaria}) = 0.2 \\ q_{\text{malaria}} &= P(\neg \text{fever} \mid \neg \text{cold}, \neg \text{flu}, \text{malaria}) = 0.1 \end{aligned}$$

If we model *fever* as a **noisy disjunction node**, then the general rule for the **CPD**

$$P(X_i \mid \text{Parents}(X_i)) = \prod_{\{j \mid X_j = T\}} q_j$$

gives the following table:

Cold	Flu	Malaria	$P(\text{Fever})$	$P(\neg \text{Fever})$
F	F	F	0.0	1.0
F	F	T	0.9	0.1
F	T	F	0.8	0.2
F	T	T	0.98	$0.02 = 0.2 \cdot 0.1$
T	F	F	0.4	0.6
T	F	T	0.94	$0.06 = 0.6 \cdot 0.1$
T	T	F	0.88	$0.12 = 0.6 \cdot 0.2$
T	T	T	0.988	$0.012 = 0.6 \cdot 0.2 \cdot 0.1$

Def 3.6. John, Mary and the Alarm: This is the well-known **Bayesian Network** example found in textbooks. It goes as follows:

- I got very valuable stuff at home. So I bought an alarm. Unfortunately, the alarm just rings at home, doesn't call me on my mobile.
- I've got two neighbors, Mary and John, who'll call me if they hear the alarm.
- The problem is that, sometimes, the alarm is caused by an earthquake.
- Also, John might confuse the alarm with his telephone, and Mary might miss the alarm altogether because she typically listens to loud music.

↪ **Random variables:** B, E, A, J, M for Burglary, Earthquake, Alarm, John, and Mary, respectively.

Given that both John and Mary call me, what is the probability of burglary?

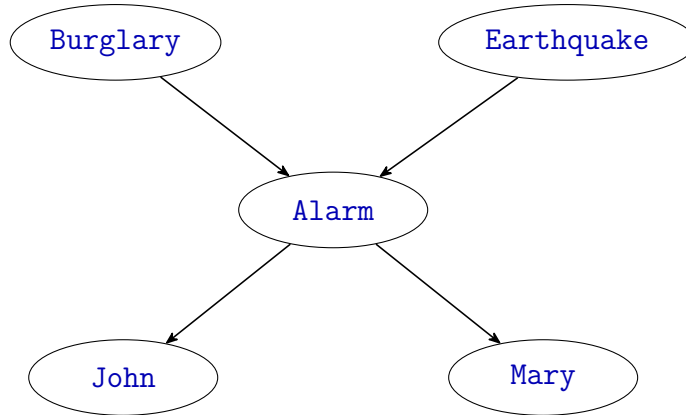


Figure 1: Burglary-Earthquake Example

Question

How to calculate $\mathbf{P}(B \mid J, M)$?

Let's use Enumeration ([Theorem 2.8](#)):

$$\mathbf{P}(Q \mid E_1 = e_1, \dots, E_{ne} = e_{ne}) = \alpha \left(\sum_{u \in D_u} \mathbf{P}(Q, E_1 = e_1, \dots, E_{ne} = e_{ne}, U_1 = u_1, \dots, U_{nu} = u_{nu}) \right)$$

with $Q = B, E = \{J, M\}, U = \{A, E\}$

$$\mathbf{P}(B \mid J, M) = \alpha \left(\sum_{b_a, b_e \in \{T, F\}} \mathbf{P}(B, J, M, A = b_a, E = b_e) \right)$$

Let's exploit [Theorem 3.1](#)

$$\mathbf{P}(X_1, \dots, X_n) = \prod_{X_i \in \mathcal{X}} \mathbf{P}(X_i \mid \text{Parents}(X_i))$$

$$\mathbf{P}(B \mid J, M) = \alpha \left(\sum_{b_a, b_e \in \{T, F\}} P(J \mid A = b_a) \cdot P(M \mid A = b_a) \cdot \mathbf{P}(A = b_a \mid E = b_e, B) \cdot P(E = b_e) \cdot \mathbf{P}(B) \right)$$

These are 5 factors in 4 summands over two cases for Burglary ↪ 38 arithmetic operations +3 for α ↪ General worst case: $\mathcal{O}(n \cdot 2^n)$

Question

Can we do better?

Answer

We can optimize a little bit:

$$\mathbf{P}(B \mid J, M) = \alpha \left(\mathbf{P}(B) \cdot \left(\sum_{b_a, b_e \in \{T, F\}} P(J \mid A = b_a) \cdot P(M \mid A = b_a) \cdot \mathbf{P}(A = b_a \mid E = b_e, B) \cdot P(E = b_e) \right) \right)$$

$$\mathbf{P}(B \mid J, M) = \alpha \left(\mathbf{P}(B) \cdot \left(\sum_{b_e \in \{T, F\}} P(E = b_e) \cdot \sum_{b_a \in \{T, F\}} \mathbf{P}(A = b_a \mid E = b_e, B) \cdot P(J \mid A = b_a) \cdot P(M \mid A = b_a) \right) \right)$$

This way, we have 3 factors in 2 summands + two factors in the outer product, over two cases $\rightsquigarrow$ 28 arithmetic operations +3 for α

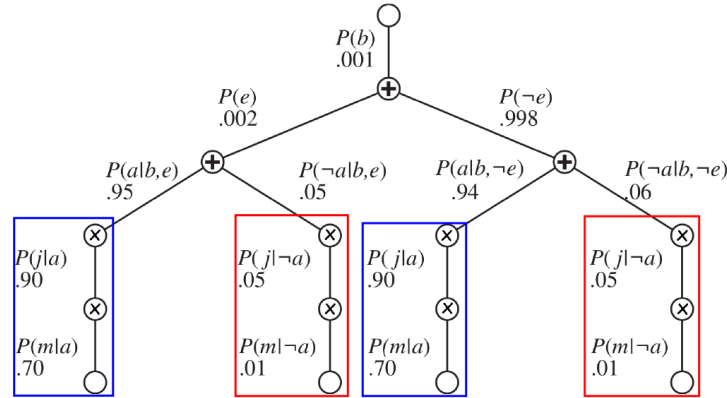
Question

Can we do better?

Answer

Yes! $\rightsquigarrow$ **Variable Elimination**

Notice that enumeration corresponds to a depth-first traversal of an arithmetic expression tree, here is an example with some example probabilities:



Notice the redundancies!

- $P(e)$ and $P(\neg e)$ (No need for both!)
- $P(j \mid \neg a), P(j \mid a), P(m \mid \neg a), P(m \mid a)$ each is being computed twice! That is because $P(J \mid A = b_a)$ and $P(M \mid A = b_a)$ only depends on A , but they are trapped behind the summation over E , hence computed twice.

We can precompute these and store them (trading time complexity by space complexity) $\rightsquigarrow$ Variable Elimination.

The idea is, instead of left-to-right (top-down DFS), we operate right-to-left (bottom-up) and store intermediate *factors* (*functions*) along with their *dependencies*

Lemma 3.2. Given a query $\mathbf{P}(Q_1, \dots, Q_{n_Q} \mid E_1 = e_1, \dots, E_{n_E} = e_{n_E})$, we can ignore and remove all hidden variables that are not ancestors of any of the $Q_1, \dots, Q_{n_Q}$ or $E_1, \dots, E_{n_E}$

A Mathematical Vernacular and Induction

A.1 Mathematical Language

Def A.1. Natural Language: A natural language is any form of spoken or signed means of communication that has evolved naturally in humans through use and repetition without conscious planning or premeditation.

Def A.2. Domain of Discourse: The [set](#) of entities, objects, or concepts that a language, discussion, or formal system refers to. It specifies the scope within which terms, statements, and reasoning apply.

Def A.3. Jargon (Terminology): A jargon (or terminology) is a [set](#) of specialized words or phrases (called technical terms or just terms) relating to concepts from a particular [domain of discourse](#).

Def A.4. Technical Language: A technical language is a [natural language](#) extended by a [terminology](#) and (possibly) special idioms, discourse markers, and notations.

Def A.5. Stylized Language: Mathematicians use a stylized language that uses formulae to represent mathematical objects, uses math idioms for special situations (e.g. "iff", "hence", "let... be...", then..."), and classifies statements by role (e.g., definition, Lemma, Theorem, Proof, Example).

Def A.6. Mathematical Vernacular: A [technical language](#) that you need to master to be successful when moving to a new environment ([symbolic AI](#)).

Def A.7. Formulae: A mathematical formula can be a **mathematical statement** (clause) which can be true or false (e.g. $x > 5, 3 + 5 = 7$), or a **mathematical object** (e.g. $3, n, x^2 + y^2 + z^2, \int_1^0 x^{3/2} dx$)

Observation: [mathematical vernacular](#) loves to name [objects/statements](#) for precise references. For example: Let $p = 3x^2 + 7x$, then $p^3 + 17p + 1$ is ...

Moreover, [mathematical vernacular](#) aggregates [objects/statements](#) for cognitive efficiency, examples:

- $a \in S, b \in S$, and $c \in S \rightsquigarrow a, b, c \in S$ ([object](#) aggregation)
- $i \geq 0$ and $i \leq n \rightsquigarrow 0 \leq i \leq n$ ([statement](#) aggregation)

Def A.8. Sequence: [mathematical vernacular](#) uses the concept of sequences instead of lists. Sequences are usually finite, but can be infinite as well.

Def A.9. Ellipses: [mathematical vernacular](#) uses ellipses ($\dots$) as a constructor for [sequences](#). The meaning of an ellipsis is usually considered "obvious" and left for interpretation by the reader.

[ellipses](#) allow to write down large [objects](#) easily, examples:

- $1, \dots, n \rightsquigarrow$ the [sequence](#) of natural numbers between 1 and n in order.
- $1, 4, 9, 16, \dots \rightsquigarrow$ the [sequence](#) of squares in orders.
- $e_1, \dots, e_n \rightsquigarrow$ a [sequence](#) of [objects](#) e_i for $1 \leq i \leq n$

[Mathematical statements](#) come in different epistemic varieties:

- **Definitions:** [statements](#) that introduce new global identifier for important [objects](#)
- **Assertions:** [statements](#) that state properties of [mathematical objects](#)
- **Examples:** [statements](#) that exhibit a witness for some property
- **Axioms:** [statements](#) that characterize the [objects](#) of a certain [domain of discourse](#) or theory. An axiom (postulate) is a statement about [mathematical objects](#) that we *assume* to be true.

[Mathematical assertions](#) are pragmatically classified into categories:

- **proofs** are arguments that justify the truth of [statements](#) beyond any doubt. [Proofs](#) are not really [statements](#), but we sometimes treat them together.
- **proposition** is a [statement](#) which is interesting in its own right.
- **lemma** is an easily proved [statement](#) which is helpful for proving other [propositions](#) and [theorems](#), but is usually not particularly interesting in its own right. A **lemma** is an intermediate [theorem](#) that serves as part of a [proof](#) of a larger [theorem](#).

- **theorem** is a more important [statement](#) than a [proposition](#) which says something definitive on the subject, and often takes more effort to prove than a [proposition](#) or [lemma](#). A [theorem](#) is a [statement](#) about [mathematical object](#) that we *know* to be true.
- **corollary** is a quick consequence of a [proposition](#) or [theorem](#) that was proven recently. It is a [theorem](#) that follows directly from another [theorem](#).
- **conjecture** is a [statement](#) that is thought to be provable, but has not been yet. A [conjecture](#) that is proven is a [theorem](#)

[Mathematical statement](#) use abbreviations, examples:

- $\wedge$ and $\vee$ for "and" and "or",
- $\neg$ for "not"
- $\forall x.P$ ($\forall x \in S.P$) for "condition P holds for all x (in S)"
- $\exists x.P$ ($\exists x \in S.P$) for "there exists an x (in S) such that the condition P holds"
- $\nexists x.P$ ($\nexists x \in S.P$) for "there exists no x (in S) such that the condition P holds"
- $\exists^1 x.P$ ($\exists^1 x \in S.P$) for "there exists one and only x (in S) such that the condition P holds"
- "*iff*" for "if and only if", symbolized by $\Leftrightarrow$
- $\Rightarrow$ for "implies"; we can read $A \Rightarrow B$ as "if A then B "

Def A.10. Declaration: In [mathematical vernacular](#) we call a clause that introduce new identifiers together with some properties a declaration. For example Let $\epsilon, \delta > 0 \dots$

Def A.11. Scope: The scope of an identifier is the part of an expression where the reference is valid; that is, where the identifier can be used to refer. In the other parts, the identifier may refer to a different entity, or to nothing at all (unbound)

Observations: [definition](#) have *global scope*, [declarations](#) have *local scope*.

Def A.12. Definiendum: The definiendum is the symbol, expression, or term being introduced by a [definition](#). It is the newly named [object](#) or concept.

Def A.13. Definiens: The definiens is the symbol, expression, or construction that specifies the meaning of the [defineiendum](#). It explains what the [defineiendum](#) denotes.

Mathematics has developed various forms of [definition](#) (definition schemata). A **simple definition** introduces a name (the [defineiendum](#)) for a compound [object](#) or concept (the [definiens](#)). The [defineiendum](#) must be new, i.e. may not have been used for anything else, in particular, the [defineiendum](#) may not occur in the [definiens](#). We use the symbols $:=$ (and the inverse $=:$) to write [simple definitions](#). For example, we can give the natural number 2 the name ϕ . In a [formula](#) we write this as $\phi := 2$ or $2 =: \phi$.

On the other hand, in a **pattern definition** the [defineiendum](#) is the operator or [relation](#) applied to n distinct variables -called pattern variables- $v_1, \dots, v_n$ as arguments, and the [definiens](#) is an expression in these variables. When the new operator is applied to arguments $a_1, \dots, a_n$, then its value is the [definiens](#) expression where the v_i are replaced by the a_i . We use the symbol $:=$ for operator definitions and $:\Leftrightarrow$ for [relation](#) definitions. For example, the following is a [pattern definition](#) for the [set intersection operator](#) $\cap$:

$$A \cap B := \{x \mid x \in A \wedge x \in B\}$$

The pattern variables are A and B , and with this definition we have e.g. $\Phi \cap \Phi = \{x \mid x \in \Phi \wedge x \in \Phi\}$.

An **implicit definition** (definition by description) is a clause or expression A , such that we can prove "there is exactly one n such that A " ($\exists^1 n.A$), where n is a new name - the [defineiendum](#). If such a unique existence proof exists, we call n *well-defined*. For example, $\forall x.x \notin \Phi$ is an [implicit definition](#) for the empty set Φ (instead of explicitly saying $\Phi := \{\}$). We can prove unique existence of Φ by just exhibiting $\{\}$ and showing that any other [set](#) S with $\forall x.x \notin S$ we have $S \equiv \Phi$. S cannot have elements, so it has the same elements as Φ , and thus $S \equiv \Phi$.

Here is another [implicit definition](#): The exponential function is that function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f' = f$ and $f(0) = 1$. Here A is the clause " $f' = f$ and $f(0) = 1$ ". Well-definedness is mathematically non-trivial.

Mathematics uses examples and counterexamples to support understanding a property P . Examples give us a sense of the extent of P and counterexamples help delineate the border of P .

Def A.14. Example: An example E is a [mathematical statement](#) that consists of:

- symbol p for the **exemplandum** (plural: exemplanda): the property to be exemplified,
- the **exemplans** (plural: exemplantia): an expression A denoting a **mathematical object** that acts as witness object for the property p , and
- (optionally) a **justification** of E , i.e. a **proof** π that $p(A)$ holds in the current context.

Def A.15. Counter Example: An **example** for the complement of the property p where π is a **proof** that $p(A)$ does not hold.

Def A.16. Running Example: An **example** that is recalled time and again during an argument, applying the newly discovered knowledge to it, presumably to show how things work.

For instance, the following **mathematical statement** is a **mathematical example**:

The identity function on $\mathbb{R}$ is *continuous*.

- the **exemplandum** is "*continuous*",
- the **exemplans** A is "The identity function on $\mathbb{R}$ ", and
- the **justification** π , a **proof** of continuity $\text{Id}_{\mathbb{R}}$: Let $\epsilon > 0$, then we choose $\delta := \epsilon \cdots$ is omitted.

We can use **examples** for $\forall$ -statements as well. Let $B = \forall x.A$ be a **mathematical statement**, then we call an **example/counter example** for B whose **exemplandum** is the property " A holds for x " an **example/counter example** for B . For example, 3 is an **example** for the **statement** $B = \forall n.\text{odd}(n)$.

Note

An **example** (or even several) for a **mathematical statement** B do not constitute a **proof** for B . On the other hand, one **counter example** for a **mathematical statement** B does show that $\neg B$ is true ($\exists x.\neg A \equiv \neg(\forall x.A)$).

Def A.17. Syllogism: A syllogism (**proof** method) is an argument that applies **deductive reasoning** to arrive at a **conclusion** based on two (or more) **propositions** that are **asserted** or assumed to be true. A **syllogistic system** is simply a **set** of **syllogisms**

Note

Today we usually reserve the term **syllogism** to informal arguments, **formal** arguments are called **inference** rules.

In a **proof by contradiction**, we make an assumption ($\neg A$) to the contrary of what we want to prove (A), and then we show that the assumption leads to a contradiction and therefore must be false. That lets us to **conclude** that $(\neg\neg A)$ must be true, and thus (A) .

If we want to prove a **statement** of the form "*If A , then B* ", then we do that by:

- assuming A and proving B from that.
- after this subproof, we may not use A anymore.

We call this **proof** method a **proof by local hypothesis**

If we know a general rule of the form "*If A then B* ", and we also know a specific instance A' , which arises from A by replacing its variables with concrete values. Then by **chaining** we can **conclude** the corresponding instance B' , obtained from B by the same variable replacement.

Example.

1. We know: "Socrates is a human." This is A' .
2. We also know a general rule:

$$\forall x (x \text{ is human} \Rightarrow x \text{ is mortal}).$$

This is the "If A then B " statement, where

$$A(x) = (x \text{ is human}), \quad B(x) = (x \text{ is mortal}).$$

3. Substitute $x = \text{Socrates}$:

$$A(x) \rightarrow \text{Socrates is human}, \quad B(x) \rightarrow \text{Socrates is mortal}.$$

4. Since “Socrates is human” is true, [chaining](#) gives us:

Socrates is mortal.

If we want to prove “ A for all x ”, then we prove A without regard for x , then argue something like “as x was chosen arbitrarily when we proved A , we know A for every x ”

If we know that one of the cases $A_1, \dots, A_k$ must always hold, then we use **proof by cases** to show that “if A_1 then C ” and “if A_2 then C ” and so on $\forall i : 1 \leq i \leq k$, hence we can [conclude](#) that C holds. The following is an example:

Lemma A.1. For all rational numbers a and b , if $ab = 0$, then $a = 0$ or $b = 0$.

Proof.

1. Let $a, b \in \mathbb{Q}$ and assume $ab = 0$.
2. Obviously, either $a = 0$ or $a \neq 0$.
3. We prove that $a = 0$ or $b = 0$ by the two cases induced.
 4. Case $a = 0$:
 - 4.1 Then the conclusion of the lemma is trivially true, and there is nothing to prove.
 5. Case $a \neq 0$:
 - 5.1 We can multiply both sides of the equation $ab = 0$ by $\frac{1}{a}$ and obtain

$$\frac{1}{a} \cdot ab = \frac{1}{a} \cdot 0.$$
 - 5.2 Reducing the fractions gives $b = 0$.
6. In both cases, either $a = 0$ or $b = 0$, so the statement holds. □

Def A.18. Without Loss of Generality (WLOG): An [inference](#) principle used in [proofs](#) when a [statement](#) $Q(x)$ must be shown for all cases of x , but the different cases are equivalent by symmetry or trivial reduction. [Formally](#), if the domain of x can be partitioned into cases $A_1, \dots, A_k$ and it can be shown that:

- proving $Q(x)$ under one representative case A_i suffices, because
- for every other case A_j there is either a trivial [proof](#) of $Q(x)$ or a direct reduction to the representative case,

then we may assume A_i holds “without loss of generality”. This allows us to simplify the argument by only proving the representative case.

For example, if we want to prove the property $Q(p)$ for all primes p . We know that every prime is either 2 (the only even prime) or an odd prime. If the case $p = 2$ is easy, and the general odd prime case is the only real work, then we can say:

[WLOG](#), assume p is odd

This does not exclude the even case; it means we have already checked that the even case adds no new difficulty.

In [mathematical vernacular](#), different alphabets are used to convey precise meanings. [Table 3](#) presents the Greek letters that you are expected to read, recognize, and write. In addition, [table 4](#) shows the curly Roman and Fraktur letters, which are also standard in mathematical writing. Becoming fluent with these alphabets is an essential part of mathematical literacy.

α	A	alpha	β	B	beta	γ	Γ	gamma
δ	Δ	delta	ϵ	E	epsilon	ζ	Z	zeta
η	H	eta	θ, ϑ	Θ	theta	ι	I	iota
κ	K	kappa	λ	Λ	lambda	μ	M	mu
ν	N	nu	ξ	Ξ	xi	$\omicron$	O	omicron
π, ϖ	Π	pi	ρ	P	rho	σ	Σ	sigma
τ	T	tau	υ	Υ	upsilon	φ, ϕ	Φ	phi
χ	X	chi	ψ	Ψ	psi	ω	Ω	omega

Table 3: Greek Alphabet

Roman	$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z}$
Fraktur	$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{O}, \mathfrak{P}, \mathfrak{Q}, \mathfrak{R}, \mathfrak{S}, \mathfrak{T}, \mathfrak{U}, \mathfrak{V}, \mathfrak{W}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$

Table 4: Roman and Fraktur Letters

A.2 Unary Natural Numbers

Def A.19. Unary Natural Numbers: Terms generated by the following: $u ::= 0 \mid s(u)$ where 0 denotes zero and $s(u)$ denotes the successor of u . i.e. $\{0, s(0), s(s(0)), s(s(s(0))), \dots\}$.

Def A.20. Piano Axioms: The following set of **axioms** are called the Piano axioms. They can be used to *reason* about natural numbers:

- **P1.** 0 is a **unary natural number**.
- **P2.** Every **unary natural number** has a successor that is **unary natural number** and that is different from it.
- **P3.** 0 is not a successor of any **unary natural number**.
- **P4.** Different **unary natural numbers** have different successors.
- **P5. Induction Axiom.** Every **unary natural number** possess a property P if
 - 0 has the property P , and
 - the successor of every **unary natural number** that has the property P also possesses property P

More formally, we call the **set** of **unary natural numbers** $\mathbb{N}_1$, and we say $P(n)$ if a **unary natural number** $n \in \mathbb{N}_1$ has a property P . Therefore, we express the axioms as follows:

- $0 \in \mathbb{N}_1$
- $\forall n \in \mathbb{N}_1. s(n) \in \mathbb{N}_1 \wedge n \neq s(n)$
- $\neg(\exists n \in \mathbb{N}_1. 0 = s(n))$
- $\forall n \in \mathbb{N}_1. \forall m \in \mathbb{N}_1. n \neq m \Rightarrow s(n) \neq s(m)$
- $\forall P. P(0) \wedge (\forall n \in \mathbb{N}_1. P(n) \Rightarrow P(s(n))) \Rightarrow (\forall m \in \mathbb{N}_1. P(m))$

Theorem A.2. Every **unary natural number** is either zero (0) or the successor of a **unary natural number**.

Proof. We make use of the induction axiom P5: We use the property P of "being zero or a successor" and prove the statement by convincing ourselves of the prerequisites of:

1. 0 is zero so 0 is "zero or a successor".
2. Let n be arbitrary **unary natural number** that is "zero or a successor".
3. Then its successor is a "successor", so the successor of n is also "zero or a successor".
4. Since we have taken n arbitrary we have shown that for any n , its successor has property P .
5. Property P holds for all **unary natural number** by P5, so we have proven the assertion.

□

Def A.21. Mathematical Induction: A method for proving a **statement** $P(n)$ is true for every natural number n . A **proof** by mathematical induction consists of two steps:

- The **base case**: Prove that $P(0)$ is true.
- The **induction step**: Prove that for every n , if $P(n)$ is true then $P(n+1)$ is also true.

The hypothesis in the **induction step**, that $P(n)$ is true for a particular n , is called the **induction hypothesis**

Def A.22. The addition "function": We define the addition operation $+$ procedurally:

- Adding 0 to a number does not change it, expressed as: $n + 0 = n$.
- Adding m to the successor of n yields the successor of $m + n$, expressed as: $m + s(n) = s(m + n)$.

Theorem A.3. For all **unary natural number** n, m, l , we have $(n + m) + l = n + (m + l)$.

Proof. We prove this by induction on l .

Let $P(l)$ be the property:

$$(n + m) + l = n + (m + l).$$

- **base case:**

$$(n + m) + 0 = (n + m) = n + (m + 0)$$

- **step case:** Assume $(n + m) + l = n + (m + l)$ for an arbitrary l . We must show

$$(n + m) + s(l) = n + (m + s(l)).$$

$$\begin{aligned} (n + m) + s(l) &= s((n + m) + l) \\ &\stackrel{\text{IH}}{=} s(n + (m + l)) \\ &= n + s(m + l) \\ &= n + (m + s(l)). \end{aligned}$$

Hence the property holds for $s(l)$.

□

Def A.23. Multiplication: The **unary multiplication** operation can be defined by the equations:

- $n \cdot 0 = 0$, and
- $n \cdot s(m) = n + (n \cdot m)$

Def A.24. Exponentiation: The **unary exponentiation** operation can be defined by the equations:

- $\text{exp}(n, 0) = s(0)$, and
- $\text{exp}(n, s(m)) = n \cdot \text{exp}(n, m)$

Def A.25. Summation: The **unary summation** operation can be defined by the equations:

- $\bigoplus_{i=0}^0 (n_i) = 0$, and
- $\bigoplus_{i=0}^{s(m)} (n_i) = n_{s(m)} \bigoplus \bigoplus_{i=0}^m (n_i)$

Example:

$$\begin{aligned} \bigoplus_{i=0}^{s(s(0))} (n_i) &= n_{s(s(0))} \bigoplus \bigoplus_{i=0}^{s(0)} (n_i) \\ &= n_{s(s(0))} \bigoplus n_{s(0)} \bigoplus \bigoplus_{i=0}^0 (n_i) \\ &= n_{s(s(0))} \bigoplus n_{s(0)} \bigoplus 0 \end{aligned}$$

Def A.26. Product: The **unary product** operation can be defined by the equations:

- $\bigodot_{i=0}^0 (n_i) = s(0)$, and
- $\bigodot_{i=0}^{s(m)} (n_i) = n_{s(m)} \bigodot \bigodot_{i=0}^m (n_i)$

B Set Theory

B.1 Sets

Def B.1. Set: A collection of different things (called **elements** or **members** of the set).

We can represent **sets** by listing the **elements** within curly brackets, e.g. $\{a, b, c\}$ or by describing the **elements** via a property, e.g. $\{x \mid x \bmod 2 = 0\}$ (the **set** of even numbers). We can state **element**-hood ($a \in S$) or not ($b \notin S$)

Note

Learn to distinguish between **objects** and their representations. ($\{a, b, c\}$ and $\{b, a, a, c\}$ are different representations of the same **set**)

Def B.2. Set Equality: $A \equiv B := \forall x. x \in A \Leftrightarrow x \in B$

Def B.3. Subset: $A \subseteq B := \forall x. x \in A \Rightarrow x \in B$

Def B.4. Proper Subset: $A \subset B := (A \subseteq B) \wedge (A \neq B)$

Def B.5. Super Set: $A \supseteq B := \forall x. x \in B \Rightarrow x \in A$

Def B.6. Proper Superset: $A \supset B := (A \supseteq B) \wedge (A \neq B)$

Def B.7. Set Union: $A \cup B := \{x \mid x \in A \vee x \in B\}$

Def B.8. Set Intersection: $A \cap B := \{x \mid x \in A \wedge x \in B\}$

Def B.9. Disjoint: Two **sets** A, B are **disjoint** iff $A \cap B = \emptyset$

Def B.10. Set Difference: $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$

Def B.11. Power Set: $\mathcal{P}(A) := \{S \mid S \subseteq A\}$ (the set of all subsets)

Note

Set operators such as **union** ($\cup$), **intersection** ($\cap$), and **set difference** ($\setminus$) always return a **set**. For example, consider

$$\{a, b\} \cap \{b, c\} = \{b\}.$$

The result $\{b\}$ is a **set**.

Now observe:

- $\{b\} \notin \{a, b\}$, since the **elements** of $\{a, b\}$ are a and b , not the **set** $\{b\}$.
- $\{b\} \notin \{b, c\}$, since the **elements** of $\{b, c\}$ are b and c , not the **set** $\{b\}$.
- $\{b\} \subseteq \{a, b\}$, because every **element** of $\{b\}$ (namely b) is also in $\{a, b\}$.
- $\{b\} \subseteq \{b, c\}$, for the same reason.

Thus, the outcome of a **set** operation should always be compared to other **sets** using $\subseteq$ (**subset relation**), not $\in$ (**element relation**).

Def B.12. Cartesian Product: $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$, (a, b) is a *pair*

Def B.13. Family: A **set** of **sets**

Def B.14. Topology: A **family** of open **sets**

Def B.15. Indexing Function: Let I and X be **sets**. A function $f : I \rightarrow X$ is called an indexing function. The set I is called the index set. For each $i \in I$, we define $x_i := f(i)$, here i is called the index (or parameter) of x_i .

Def B.16. Indexed Family: Given an **indexing function** $f : I \rightarrow X$, the **set** of values $f(I) = \{f(i) : i \in I\}$ is called an indexed **family**. It is often written as $(x_i)_{i \in I}$, $\langle x_i \rangle_{i \in I}$, or $\{x_i\}_{i \in I}$.

Example: Consider **index set** $I := \{1, 2, 3\}$. Consider **indexing function** f defined as:

$$f(1) = \{a, b\} \quad f(2) = \{c\} \quad f(3) = \{d, f\}$$

the **set** of values $f(I) = \{f(1), f(2), f(3)\}$ is the **indexed family**:

$$(X_i)_{i \in I} = f(I) = \{\{a, b\}, \{c\}, \{d, f\}\}$$

where $x_i = f(i)$, i.e. $x_1 = f(1) = \{a, b\}, \dots$

Def B.17. Union over a Family: Let $(X_i)_{i \in I}$ be an **indexed family**, then $\bigcup_{i \in I} X_i := \{x \mid \exists i \in I. x \in X_i\}$

Def B.18. Intersection over a Family: Let $(X_i)_{i \in I}$ be an **indexed family**, then $\bigcap_{i \in I} X_i := \{x \mid \forall i \in I. x \in X_i\}$

Def B.19. n -fold Cartesian Product: $\prod_{i=1}^n X_i := X_1 \times \dots \times X_n := \{\langle x_1, \dots, x_n \rangle \mid \forall i. 1 \leq i \leq n \Rightarrow x_i \in X_i\}$ where $\langle x_1, \dots, x_n \rangle$ is called an n -tuple.

Example: consider the **indexed family** $(X_i)_{i \in \{1,2,3\}} = \{\{a, b\}, \{c\}, \{d, f\}\}$. $X_1 \times X_2 \times X_3 = \{\langle a, c, d \rangle, \langle a, c, f \rangle, \langle b, c, d \rangle, \langle b, c, f \rangle\}$.

Def B.20. n -dim Cartesian Space: $X^n := \{\langle x_1, \dots, x_n \rangle \mid 1 \leq i \leq n \Rightarrow x_i \in X\}$ where $\langle x_1, \dots, x_n \rangle$ is called a vector.

Example: Let $X = \{1, 2\}$, then $X^2 = \{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 2, 1 \rangle, \langle 2, 2 \rangle\}$. Another example is the usual 2D plane: $\mathbb{R}^2 := \{\langle x, y \rangle \mid x \in \mathbb{R}, y \in \mathbb{R}\}$

Def B.21. Size of a Set: The size $\Gamma(A)$ of a **set** A is the number of **elements** in A .

Note

- $\Gamma(A \cup B) \leq \Gamma(A) + \Gamma(B)$
- $\Gamma(A \cap B) \leq \min(\Gamma(A), \Gamma(B))$
- $\Gamma(A \times B) = \Gamma(A) \cdot \Gamma(B)$

Def B.22. Pairwise Disjoint: A **family** of **sets** is called **pairwise disjoint** or **mutually disjoint**, if any two of those **sets** are **disjoint**.

Def B.23. Multiset: A **multiset** (or **bag**) is a generalization of a **set** where **elements** can appear more than once. Formally, a multiset $\mathcal{M}$ over a domain D is a pair (D, m) , where $m : D \rightarrow \mathbb{N}$ is a function mapping each **element** $x \in D$ to its **multiplicity** (the number of occurrences).

Def B.24. Multiset Membership: $x \in \mathcal{M} \equiv m(x) > 0$

Def B.25. Multiset Size: The size of a multiset $\mathcal{M} = (D, m)$, denoted $\Gamma(\mathcal{M})$, is the sum of the multiplicities of all its **elements**: $\Gamma(\mathcal{M}) = \sum_{x \in D} m(x)$.

Def B.26. Multiset Sum (Union): The sum of two multisets $\mathcal{M}_1 = (D, m_1)$ and $\mathcal{M}_2 = (D, m_2)$, denoted $\mathcal{M}_1 \uplus \mathcal{M}_2$, is a multiset where the multiplicity of each **element** is the sum of its multiplicities: $m_{\uplus}(x) = m_1(x) + m_2(x)$.

Def B.27. Multiset Intersection: The **intersection** of two multisets $\mathcal{M}_1 = (D, m_1)$ and $\mathcal{M}_2 = (D, m_2)$, denoted $\mathcal{M}_1 \cap \mathcal{M}_2$, is a multiset where the multiplicity of each **element** is the minimum of its multiplicities in the two multisets: $m_{\cap}(x) = \min(m_1(x), m_2(x))$.

Def B.28. Multiset Equality: $\mathcal{M}_1 \equiv \mathcal{M}_2 \equiv \forall x \in D. m_1(x) = m_2(x)$

B.2 Relations

Def B.29. Relation: $R \subseteq A \times B$ is a (binary) relation between A and B

Def B.30. Relation on: a relation $R \subseteq A \times B$ where $A = B$ is called a *relation on* A

Def B.31. Total: A relation $R \subseteq A \times B$ is called *total* (*left total*) iff $\forall x \in A. \exists y \in B. (x, y) \in R$

Def B.32. Converse Relation: $R^{-1} \subseteq B \times A := \{(y, x) \mid (x, y) \in R\}$ is the converse relation of R

Def B.33. Relation Composition: The composition of $R \subseteq A \times B$ and $S \subseteq B \times C$ is defined as $R \circ S := \{(a, c) \in A \times C \mid \exists b \in B. (a, b) \in R \wedge (b, c) \in S\}$

A relation on A : $R \subseteq A \times A$ is called:

- **reflexive** on A , iff $\forall a \in A. (a, a) \in R$
- **irreflexive** on A , iff $\forall a \in A. (a, a) \notin R$
- **symmetric** on A , iff $\forall a, b \in A. (a, b) \in R \Rightarrow (b, a) \in R$
- **asymmetric** on A , iff $\forall a, b \in A. (a, b) \in R \Rightarrow (b, a) \notin R$
- **antisymmetric** on A , iff $\forall a, b \in A. (a, b) \in R \wedge (b, a) \in R \Rightarrow a = b$
- **transitive** on A , iff $\forall a, b, c \in A. (a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R$
- **equivalence relation** on A , iff R is reflexive symmetric, and transitive

Def B.34. Equality Relation: The equality relation is an equivalence relation on any set.

Def B.35. Divides Relation: The *divides* relation on the integers is defined as $\mid \subseteq \mathbb{Z} \times \mathbb{Z}$, where $\mid = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid \exists k \in \mathbb{Z} : b = a \cdot k\}$. We write $a \mid b$ to denote $(a, b) \in \mid$ (read as "*a divides b*").

For example: $5 \mid 10$ because $\exists k : 10 = 5 \cdot k$ where $k = 2$.

Def B.36. Congruence Modulo: For a fixed $n \in \mathbb{N}$ with $n \geq 1$, the *congruence modulo* n relation on the integers is defined as

$$\equiv_n \subseteq \mathbb{Z} \times \mathbb{Z}, \quad \equiv_n = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid n \mid (a - b)\}.$$

We write $a \equiv b \pmod{n}$ to denote $(a, b) \in \equiv_n$ (read as "*a is congruent to b modulo n*").

Examples:

- $12 \equiv 2 \pmod{5}$ because $5 \mid 10$
- $7 \equiv 22 \pmod{5}$ because $5 \mid 15$
- $9 \equiv 9 \pmod{5}$ because $5 \mid 0$

Note

In the definition of congruence modulo n , we do *not* take the absolute value of the difference:

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Since the divisor n can be multiplied by a negative integer, the sign of $(a - b)$ does not matter. For example, $2 \equiv 12 \pmod{5}$ because $2 - 12 = -10$ and $-10 = 5 \cdot (-2)$, so $5 \mid (2 - 12)$.

Exercise: Which of the following binary relations on integers are equivalence relations?

1. Φ . It is not reflexive because we must have $(a, a) \in R$ for all integers a . But Φ has no pairs at all. $\rightarrow$ NO.
2. $\mathbb{Z} \times \mathbb{Z}$
 - It is reflexive since $(a, a) \in R \quad \forall a$
 - It is symmetric because if $(a, b) \in R$ then (b, a) also is
 - It is transitive because every possible pair is in it anyway
 - $\rightarrow$ YES.
3. $m \leq n$
 - It is reflexive since $m \leq m$

- It is not **symmetric**, e.g. $2 \leq 3$ but not $3 \leq 2$

- $\rightarrow$ NO

4. $\exists z \in \mathbb{Z} : m + z = n$

- This is actually the same as $\mathbb{Z} \times \mathbb{Z}$ because for any integers m, n , we can always pick $z = n - m$. $\rightarrow$ YES.

5. $\exists z \in \mathbb{Z} : m \cdot z = n$

- This is the same as $m \mid n$. It is **reflexive** ($m \cdot 1 = m$). However, it is not **symmetric** ($2 \mid 4$ but $4 \nmid 2$). $\rightarrow$ NO.

6. $5 \mid (m - 2)$

- This means m is **congruent** to 2 modulo 5.
- It is **reflexive** ($5 \mid 0$)
- It is **symmetric** (negative or positive does not matter)
- It is **transitive**, if $5 \mid m - n$ and $5 \mid n - k$, then $5 \mid (m - n) + (n - k) = m - k$
- $\rightarrow$ YES.

7. $m \mid n$. It is **reflexive** but not **symmetric**. $\rightarrow$ NO.

Def B.37. Reflexive Closure: The **reflexive closure** of a **relation** R on a **set** A is the **relation** $R \cup \Delta$ where $\Delta = \{\langle a, a \rangle \mid a \in A\}$ is the identity relation. It is defined as:

$$R_r = R \cup \{\langle a, a \rangle \mid a \in A\}$$

Note

Alternative notation: $R_r = R \cup R^0$. Since R^n is the **set** of pairs that are connected by a chain of exactly n hops "reachable in n steps", we can think of R^0 as pairs that are connected by a chain of exactly 0 hops "reachable in 0 steps", i.e. the identity relation.

For example, let $A = \{1, 2, 3\}$, $R = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle\}$, then $R_r = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle\}$

Def B.38. Symmetric Closure: The **symmetric closure** of a **relation** R on a **set** A is the **union** of R and its inverse **relation** R^{-1} . It is defined as:

$$R_s = R \cup R^{-1} = R \cup \{\langle b, a \rangle \mid \langle a, b \rangle \in R\}$$

e.g. $R_s = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 2, 1 \rangle, \langle 3, 2 \rangle\}$

Def B.39. Transitive Closure: The **transitive closure** R^+ of a **relation** R on a **set** A is the **smallest transitive relation** containing R . It is defined as:

$$R^+ = \bigcup_{n=1}^{\infty} R^n$$

For example, let $A = \{1, 2, 3, 4\}$, $R = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle\}$

- $R^1 = R$
- $R^2 = R \circ R = \{\langle 1, 3 \rangle, \langle 2, 4 \rangle\}$
- $R^3 = R \circ R \circ R = \{\langle 1, 4 \rangle\}$
- No further compositions yields new pairs.

$\rightsquigarrow R^+ = R \cup R^2 \cup R^3 = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle, \langle 1, 3 \rangle, \langle 2, 4 \rangle, \langle 1, 4 \rangle\}$

Def B.40. Reflexive Transitive Closure: The **reflexive transitive closure** (A.K.A ??) of a **relation** R on a **set** A is the **smallest relation** that is both **reflexive** and **transitive**. It is defined as:

$$R^* = R^0 \cup R^+ = \bigcup_{n=0}^{\infty} R^n$$

Example: let $A = \{1, 2, 3\}$, $R = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle\}$.

- R^0 is the identity relation on $A = \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle\}$
- $R^1 = R$
- $R^2 = \{\langle 1, 3 \rangle\}$
- No new pairs can be introduced

$$\rightsquigarrow R^* = \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle\} \cup \{\langle 1, 2 \rangle, \langle 2, 3 \rangle\} \cup \{\langle 1, 3 \rangle\} = \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle\}$$

Def B.41. Parital Ordering (non-strict): A relation $R \subseteq A \times A$ is called a **partial ordering** on A , iff R is **reflexive**, **antisymmetric**, and **transitive** on A .

Def B.42. Strict Partial Ordering: A relation $R \subseteq A \times A$ is called a **strict partial ordering** on A , iff R is **irreflexive**, and **transitive** on A .

Note

We usually write **strict partial ordering** with asymmetric symbols like $\prec$ and **non-strict ones** by adding a line that reminds us of equality ($\preceq$).

Def B.43. Comparable: In a **non-strict partial ordering**, two elements a, b are comparable if either $a \preceq b$ or $b \preceq a$. If neither holds, they are incomparable.

Def B.44. Linear Order: A **partial ordering** is called linear (or total) on A , iff all **elements** in A are **comparable**, i.e. if $(x, y) \in R$ or $(y, x) \in R$ for all $x, y \in A$.

For example, the $\leq$ relation is a **linear order** on $\mathbb{N}$. However, the "divides" ($\mid$) relation is a **non-strict partial ordering** on $\mathbb{N}$ that is not **linear** (e.g., neither $2 \mid 3$ nor $3 \mid 2$).

Lemma B.1. **Strict partial orderings** are **asymmetric**.

Proof. By contradiction: if $(a, b) \in R$ and $(b, a) \in R$, then $(a, a) \in R$ by **transitivity**, and that can't be because its **irreflexive**. $\square$

Lemma B.2. If $\preceq$ is a **non-strict partial order**, then $\prec := \{(a, b) \mid a \preceq b \wedge a \neq b\}$ is a **strict partial order**. Conversely, if $\prec$ is a **strict partial order**, then $\preceq := \{(a, b) \mid a \prec b \vee a = b\}$ is a **non-strict partial order**.

Proof. (1) **Non-strict** $\rightarrow$ **Strict**: Assume $\preceq$ is reflexive, antisymmetric, and transitive. Define $a \prec b \iff (a \preceq b \wedge a \neq b)$.

- *Irreflexive:* $a \prec a$ would require $a \neq a$, impossible.
- *Transitive:* If $a \prec b$ and $b \prec c$, then $a \preceq b$, $b \preceq c$, and $a \neq b$, $b \neq c$. By transitivity of $\preceq$, $a \preceq c$. If $a = c$, antisymmetry would force $a = b$, contradiction. So $a \neq c$, hence $a \prec c$.

Thus $\prec$ is a strict partial order.

(2) **Strict** $\rightarrow$ **Non-strict**: Assume $\prec$ is irreflexive and transitive. Define $a \preceq b \iff (a = b \vee a \prec b)$.

- *Reflexive:* $a = a$ gives $a \preceq a$.
- *Antisymmetric:* If $a \preceq b$ and $b \preceq a$, then either $a = b$ or we would have $a \prec b$ and $b \prec a$, contradicting asymmetry.
- *Transitive:* If $a \preceq b$ and $b \preceq c$, then combining cases ($=$ or $\prec$) always gives $a \preceq c$.

Thus $\preceq$ is a non-strict partial order. $\square$

Examples:

- On $\mathbb{N}$, the non-strict partial order $\leq$ induces the strict order $<$ by removing equality. Conversely, $<$ induces $\leq$ by adding equality.
- On sets, the non-strict partial order $\subseteq$ induces the strict order $\subset$. Conversely, $\subset$ induces $\subseteq$ by adding equality.

Exercise: which of the following binary relation on integers are **partial ordering**:

1. Φ . It is not **reflexive** because we must have $(a, a) \in R$ for all integers a . But Φ has no pairs at all. $\rightarrow$ NO.
2. $\mathbb{Z} \times \mathbb{Z}$
 - It is **reflexive** since $(a, a) \in R \quad \forall a$
 - It is **symmetric** because if $(a, b) \in R$ then (b, a) also is, $\rightarrow$ it is not **antisymmetric**, $\rightarrow$ NO
3. $m \leq n$

- It is **reflexive** since $m \leq m$
- It is **asymmetric**, e.g. $2 \leq 3$ but not $3 \leq 2$
- It is **transitive**
- $\rightarrow$ YES

4. $\exists z \in \mathbb{Z} : m + z = n$

- This is actually the same as $\mathbb{Z} \times \mathbb{Z}$ because for any integers m, n , we can always pick $z = n - m$. $\rightarrow$ NO.

5. $\exists z \in \mathbb{Z} : m \cdot z = n$

- This is the same as $m \mid n$. It is **reflexive** ($m \cdot 1 = m$). It is **asymmetric** on $\mathbb{N}$ but not on $\mathbb{Z}$ ($2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$). $\rightarrow$ NO.

6. $5 \mid (m - 2)$

- This means m is **congruent** to 2 modulo 5 .
- It is **reflexive** ($5 \mid 0$)
- It is **symmetric** (not **asymmetric**)
- It is **transitive**, if $5 \mid m - n$ and $5 \mid n - k$, then $5 \mid (m - n) + (n - k) = m - k$
- $\rightarrow$ NO.

7. $m \mid n$. It is **reflexive** but not **asymmetric**. $\rightarrow$ NO.

Def B.45. Partially Ordered Set (Poset): A **Partially Ordered Set** (or **poset**) is a pair $\langle G, \preceq \rangle$ where G is a **set** and $\preceq$ is a **non-strict partial ordering** on G .

Def B.46. Least Element: Let $\langle G, \preceq \rangle$ be a **poset** and $T \subseteq G$. An **element** $t \in T$ is the **least** (or **smallest**, **minimal**, or **$\preceq$ -minimal**) element of T iff $t \preceq t'$ for all $t' \in T$.

Def B.47. Greatest Element: Let $\langle G, \preceq \rangle$ be a **poset** and $T \subseteq G$. An **element** $t \in T$ is the **greatest** (or **largest**, **maximal**, or **$\preceq$ -maximal**) element of T iff $t' \preceq t$ for all $t' \in T$.

Def B.48. Supremum and Infimum: Let $\langle G, \preceq \rangle$ be a **non-strict partial ordering** and $T \subseteq G$, then we call the **least upper bound** ($\sup(T) \in G$) of T the **supremum** of T and the **greatest lower bound** ($\inf(T) \in G$) of T the **infimum** of T (if they exist).

Example: Consider the **non-strict partial ordering** $\langle \mathbb{R}, \preceq \rangle$. Let T be the open interval $(0, 1) \subseteq \mathbb{R}$, then $\sup(T) = 1$, and $\inf(T) = 0$. Note that $0, 1 \in \mathbb{R}$ but $\notin T$. (its different from **min** and **max** of T).

B.3 Functions

Def B.49. Partial function: A [relation](#) $R \subseteq X \times Y$ is a *partial function* from X to Y ($f : X \rightarrow Y$) iff $\forall x \in X, \forall y_1, y_2 \in Y. ((x, y_1) \in R \wedge (x, y_2) \in R \Rightarrow y_1 = y_2)$ (i.e. there is at most one $y \in Y$ with $(x, y) \in f$)

Def B.50. dom, codom: We call X the domain ($\text{dom}(f)$), and Y the codomain ($\text{codom}(f)$)

Def B.51. Function Application: Instead of writing $(x, y) \in f$ we write $f(x) = y$

Def B.52. Undefined: we call a [partial function](#) $f : X \rightarrow Y$ undefined at $x \in X$, iff $\forall y \in Y. (x, y) \notin f$ ($f(x) = \perp$).

Def B.53. Function: A [partial function](#) $f : X \rightarrow Y$ is called a *function* (or *total function*) from X to Y iff it is a [total relation](#). ($\forall x \in X. \exists^1 y \in Y : (x, y) \in f$)

Note

A [partial function](#) guarantees uniqueness, but not existence. So each x has at most one y (some might have none). A [total function](#) guarantees both uniqueness and existence (at most and at least $\rightarrow$ exactly one)

Def B.54. Partial Function Space: Given two sets X and Y , the [set](#) of all possible [partial functions](#) from X to Y is called the partial function space, denoted as $X \rightarrow Y$

Note

If X and Y are finite, then for each element $x \in X$ we have $\Gamma(Y) + 1$ choices (either pick one of the $\Gamma(Y)$ values in Y , or leave it [undefined](#)). Therefore, $\Gamma(X \rightarrow Y) = (\Gamma(Y) + 1)^{\Gamma(X)}$.

Example, $X = Y = \{0, 1\}$

$$X \rightarrow Y = \{\Phi, \{(0, 0)\}, \{(0, 1)\}, \{(1, 0)\}, \{(1, 1)\}, \{(0, 0), (1, 0)\}, \{(0, 1), (1, 1)\}, \{(0, 1), (1, 0)\}, \{(0, 0), (1, 1)\}\}$$

Notice that $\Gamma(X \rightarrow Y) = 9 = (\Gamma(Y) + 1)^{\Gamma(X)} = (2 + 1)^2$

Def B.55. Function Space: Given two sets X and Y , the [set](#) of all possible [functions](#) from X to Y is called the partial function space, denoted as $X \rightarrow Y$

Note

If X and Y are finite, then for each element $x \in X$ we must pick exactly one of the $\Gamma(Y)$ values in Y . Therefore, $\Gamma(X \rightarrow Y) = (\Gamma(Y))^{\Gamma(X)}$.

Example, $X = Y = \{0, 1\}$

$$X \rightarrow Y = \{\{(0, 0), (1, 0)\}, \{(0, 1), (1, 1)\}, \{(0, 1), (1, 0)\}, \{(0, 0), (1, 1)\}\}$$

Notice that $\Gamma(X \rightarrow Y) = 4 = (\Gamma(Y))^{\Gamma(X)} = (2)^2$

A [function](#) $f : X \rightarrow Y$ is called

- **injective** iff $\forall x_1, x_2 \in X. f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
- **surjective** iff $\forall y \in Y. \exists x \in X. f(x) = y$
- **bijective** iff f is [injective](#) and [surjective](#)

Let's look for some examples to make the previous definitions clear.

Assume $X = \{1, 2, 3\}$ $Y = \{a, b\}$ The [relation](#) $R = \{(1, a), (1, b)\}$ is just a [relation](#), it is not a [partial function](#) nor a [total function](#).

If $R = \{(1, a)\}$ it becomes a [partial function](#) but not a [total one](#).

If $R = \{(1, a), (2, a), (3, a)\}$ it becomes a [total function](#) because each $x \in X$ has exactly one $y \in Y$. However, it is not [injective](#) (1, 2, 3 all map to the same value) and not [surjective](#) (b is not reached).

IF $R = \{(1, a), (2, a), (3, b)\}$ then this is a [surjective function](#) but not an [injective one](#) (since 1, 2 map to the same value).

Assume $X = \{1, 2\}$ $Y = \{a, b, c\}$ $f = \{(1, a), (2, b)\}$ then this is an [injective function](#) (different inputs map to different output) but not [surjective](#) since there is no x such that $f(x) = c$.

And finally, assume $X = \{1, 2\}$, $Y = \{a, b\}$, $f = \{(1, a), (2, b)\}$ then this is a **total function** that is both **injective** and **surjective**, i.e. a **bijective function**.

Def B.56. Function Composition: If $f : A \rightarrow B$ and $g : B \rightarrow C$ are **functions**, then we call $g \circ f : A \rightarrow C; x \mapsto g(f(x))$ the composition of g and f (read g after f).

Note

Given $f, g : \mathbb{N} \rightarrow \mathbb{N}$, if f and g both are **injective/bijective**, so is $g \circ f$

Observations:

- If f is **injective**, then the **converse relation** f^{-1} is a **partial function**.

For example: $X = \{1, 2, 3\}, Y = \{a, b, c, d\}, f : X \rightarrow Y = \{(1, a), (2, b), (3, c)\}$. f is clearly **injective** since different inputs map to different outputs. $\text{dom}(f^{-1}) = \text{codom}(f)$ and vice versa. $f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$. It is a **partial function** because for $d \in \text{dom}(f^{-1})$ there is no $f^{-1}(d)$

- If f is **surjective**, then the **converse relation** f^{-1} is a **total relation**.

For example: $X = \{1, 2, 3\}, Y = \{a, b\}, f : X \rightarrow Y = \{(1, a), (2, b), (3, b)\}$. f is clearly **surjective** because both a, b are hit by some x . $f^{-1} = \{(a, 1), (b, 2), (b, 3)\}$. It is a **total function**.

- If f is **bijective**, call the **converse relation** **inverse function**, we also write it as f^{-1} .

- If f is **bijective**, then the **inverse function** f^{-1} is a **total function**.

- If $f : X \rightarrow Y$ is **bijective**, then $f \circ f^{-1} = \text{Id}_Y$.

For example: $X = \{1, 2, 3\}, Y = \{a, b, c\}, f = \{(1, a), (2, b), (3, c)\}, f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$. $(f \circ f^{-1})(y) = f(f^{-1}(y))$

$$- y = a : f^{-1}(a) = 1, f(1) = a$$

$$- y = b : f^{-1}(b) = 2, f(2) = b$$

$$- y = c : f^{-1}(c) = 3, f(3) = c$$

$$\rightarrow f \circ f^{-1} = \text{Id}_Y$$

Def B.57. Image and Preimage: Let $f : X \rightarrow Y$ be a **function**, $X' \subseteq X$ and $Y' \subseteq Y$, then we call

- $f(X') := \{y \in Y \mid \exists x \in X'. (x, y) \in f\}$ the **image** of X' under f ,
- $\text{Im}(f) := f(X)$ the **image** of f , and
- $f^{-1}(Y') := \{x \in X \mid \exists y \in Y'. (x, y) \in f\}$ the **preimage** of Y' under f .

Note

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be **injective** and let its **partial inverse** be

$$f^{-1} : \mathbb{N} \rightarrow \mathbb{N}, \quad \text{dom}(f^{-1}) = \text{im}(f), \quad f^{-1}(y) = x \text{ iff } f(x) = y.$$

Then $f^{-1} \circ f = \text{Id}_{\mathbb{N}}$ as a **total function** $\mathbb{N} \rightarrow \mathbb{N}$.

Proof. For any $x \in \mathbb{N}$ we have $f(x) \in \text{im}(f) = \text{dom}(f^{-1})$, hence $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x$. $\square$

For example: Define $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = 2x$. Then f is **injective** with **image** $\text{im}(f) = \{0, 2, 4, 6, \dots\}$. Its **partial inverse** is

$$f^{-1} : \mathbb{N} \rightarrow \mathbb{N}, \quad f^{-1}(y) = \frac{y}{2} \text{ if } y \text{ is even, undefined if } y \text{ is odd.}$$

For all $x \in \mathbb{N}$ we have

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(2x) = \frac{2x}{2} = x,$$

hence $f^{-1} \circ f = \text{Id}_{\mathbb{N}}$. By contrast, $f \circ f^{-1}$ is only the identity on the even numbers.

Another example: Define $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = x + 1$. Then f is **injective** with **image** $\text{im}(f) = \{1, 2, 3, \dots\}$. Its **partial inverse** is

$$f^{-1} : \mathbb{N} \rightarrow \mathbb{N}, \quad f^{-1}(y) = y - 1 \text{ if } y \geq 1, \quad \text{undefined if } y = 0.$$

For all $x \in \mathbb{N}$ we have

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(x+1) = (x+1) - 1 = x,$$

hence again $f^{-1} \circ f = \text{Id}_{\mathbb{N}}$. Here $f \circ f^{-1}$ is only the identity on $\{1, 2, 3, \dots\}$.

Def B.58. Isomorphic: We call two **sets** A and B isomorphic iff there is a **bijection** $f : A \rightarrow B$.

Def B.59. Cardinality: We say that a **set** A is **finite** and has **cardinality** $\Gamma(A) \in \mathbb{N}$, iff there is a **bijjective function** $f : A \rightarrow \{n \in \mathbb{N} \mid n < \Gamma(A)\}$.

Def B.60. Countably Infinite: We say that a **set** A is countably infinite, iff there is a **bijjective function** $f : A \rightarrow \mathbb{N}$. The **cardinality** of such set is denoted by $\aleph_0$. Examples: $\mathbb{N}, \mathbb{Z}, \dots$

Def B.61. Uncountably Infinite: We say that a **set** A is **uncountably infinite**, iff A is infinite and there is **no bijjective function** $f : A \rightarrow \mathbb{N}$. The **cardinality** of the such sets is denoted by $\mathfrak{c}$ (for continuum). Examples: $\mathbb{R}, [0, 1], \mathbb{C}, \mathcal{P}(\mathbb{N})$

Def B.62. Countable: A **set** A is called countable, iff it is **finite** or **countably infinite**

Note

The **set** of real numbers $\mathbb{R}$ is not **countable** (there is no **bijection**) $\mathbb{R} \rightarrow \mathbb{N}$

Def B.63. Restriction: Let $f : A \rightarrow B$ and $C \subseteq A$, then we call the **function** $f \upharpoonright_C := \{(c, b) \in f \mid c \in C\}$ the restriction of f to C .

Def B.64. Lambda-Notation: Let X be a **set** and E an expression depending on a variable $x \in X$. The expression $\lambda x \in X. E$ denotes the (**partial**) **function** f from X to Y with $f(x) = E$ for all $x \in X$.

Examples:

- the identity function is written in **lambda notation**: $\lambda n \in \mathbb{N}. n = \{(n, n) \mid n \in \mathbb{N}\} = \text{Id}_{\mathbb{N}} = \{(0, 0), (1, 1), (2, 2), \dots\}$
- the square function: $\lambda x \in \mathbb{N}. x^2 = \{(x, x^2) \mid x \in \mathbb{N}\} = \{(0, 0), (1, 1), (2, 4), (3, 9), \dots\}$
- the "add" operation: $\lambda(x, y) \in \mathbb{N} \times \mathbb{N}. x + y = \{((0, 0), 0), ((1, 0), 1), \dots\}$

Def B.65. Curried Function Type: Let X, Y, Z be sets. An *uncurried* **function** is written as

$$f : X \times Y \rightarrow Z, \quad f(x, y) = E(x, y).$$

The same **function** can equivalently be written in *curried* form as

$$f : X \rightarrow Y \rightarrow Z, \quad f(x)(y) = E(x, y).$$

Thus in the curried notation the **function** takes one argument at a time: for $x \in X$, the value $f(x)$ is itself a **function** $Y \rightarrow Z$, and applying it to $y \in Y$ yields $f(x)(y) \in Z$.

Example. The addition **function** can be written in **uncurried** or **curried** form:

$$\text{add} : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, \quad \text{add}(m, n) = m + n,$$

equivalently: $\text{add} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad \text{add}(m)(n) = m + n.$

In the **curried** form, for fixed $m \in \mathbb{N}$, the expression $\text{add}(m) : \mathbb{N} \rightarrow \mathbb{N}$ is the **function** $n \mapsto m + n$, i.e. the **function** that adds m to its argument.

Another Example (Ternary)

$$h : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, \quad h(m, n, o) = m + n + o.$$

Equivalently, in **curried form** we can write

$$h : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad h(m)(n)(o) = m + n + o.$$

In this form, $h(m)$ is a **function**

$$h(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad h(m)(n)(o) = m + n + o,$$

which for fixed m returns the **function** "add m to the sum of two further arguments."

Def B.66. Well-Defined Function: Let X, Y be sets. A **function** declaration $f : X \rightarrow Y$ together with a defining expression $E(x)$ is called *well-defined* iff for every $x \in X$:

- the expression $E(x)$ is meaningful (all variables are bound and the operations are valid), and
- the result $E(x)$ lies in Y .

Equivalently: a **function** is well-defined if its type annotation $f : X \rightarrow Y$ is consistent with the expression that defines it.

Checking well-definedness in practice:

The general definition above says that a **function** declaration must be consistent with its defining expression. Typical ways in which this can fail are:

- **Type mismatch:** the declared type $X \rightarrow Y$ does not agree with the actual input/output structure of the definition.
- **Free variables:** the definition uses symbols that are not bound by the input.
- **Partiality:** the definition does not provide an output for all inputs in the domain.

To illustrate, let us fix two **functions**

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, \quad g : \mathbb{N} \rightarrow \mathbb{N},$$

and examine a series of possible definitions of **functions** h . Each case shows whether the definition is **well-defined** and why.

A correct curried definition

$$h : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad h(m)(n) = f(n, g(m)).$$

This is well-defined: for each $m \in \mathbb{N}$ we obtain $g(m) \in \mathbb{N}$, so $(n, g(m)) \in \mathbb{N} \times \mathbb{N}$ and $f(n, g(m)) \in \mathbb{N}$.

Incorrect composition

$$g \circ f : \mathbb{N} \rightarrow \mathbb{N}.$$

This is not well-defined with the given type. Since $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, the composition $g \circ f$ has domain $\mathbb{N} \times \mathbb{N}$, hence $g \circ f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, not $\mathbb{N} \rightarrow \mathbb{N}$.

A correct nested definition

$$h : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad h(m)(n, o) = f(n, f(m, o)).$$

This is **well-defined**: $f(m, o) \in \mathbb{N}$, so $(n, f(m, o)) \in \mathbb{N} \times \mathbb{N}$, and thus $f(n, f(m, o)) \in \mathbb{N}$.

Type mismatch between **curried** and **uncurried**

$$h : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}, \quad h = f.$$

This is not **well-defined**, since $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ has a different type from the declared type of h . Mathematically $f(x, y)$ and $f(x)(y)$ can be identified via **currying**, but in type-theoretic terms they are distinct unless an explicit **curry** operation is applied.

Free variable problem

$$h : \mathbb{N} \rightarrow \mathbb{N}, \quad h(x) = f(x, y).$$

This is not **well-defined**, since y is not bound. Without fixing y in advance, the right-hand side is not a **function** of x alone.

Partial definition problem

$$h : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, \quad h(x, 1) = x.$$

This is not a **total function**, because the definition only specifies values when the second argument is 1. A **function** of type $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ must assign a value for every **pair** (x, y) .

Def B.67. Invertible: Let A and B be **sets** and $f : A \rightarrow B$ a **function**, then f is called invertible, iff there is a **function** $g : B \rightarrow A$, such that $f \circ g = \text{Id}_B$. g is called the **inverse function** (or just **inverse**) of f and is written as f^{-1} .

Theorem B.3. A **function** $f : A \rightarrow B$ is **invertible**, iff it is **bijective**.

Proof.

1. If f is **bijective**, then it is **invertible**:

- Assume $f : A \rightarrow B$ is **bijective**
- for each $b \in B$ there exists a unique $a \in A$ such that $f(a) = b$. Define $g : B \rightarrow A$ by sending b to this unique a . Therefore:
 1. $g \circ f = \text{Id}_A$, because if you start with $a \in A$, apply f , then g , you get back a .
 2. $f \circ g = \text{Id}_B$, because if you start with $b \in B$, apply g , then f , you get back b .

- Hence, g is the **inverse** of f . So f is **invertible**

2. If f is **invertible**, then it is **bijjective**:

Suppose there exists a **function** $g : B \rightarrow A$ such that $g \circ f = \text{Id}_A$ and $f \circ g = \text{Id}_B$. We want to show f is **bijjective**.

1. **injectivity**

(a) Assume $f(a) = f(a')$.

(b) Apply g to both sides:

$$a = \text{Id}_A(a) = g(f(a)) = g(f(a')) = a'$$

(c) So, $a = a'$. Hence f is **injective**.

2. **surjectivity**

(a) Take any $b \in B$. Then

$$f(g(b)) = (f \circ g)(b) = \text{Id}_B(b) = b$$

(b) So b is indeed an image of f . Hence f is **surjective**.

□

Lemma B.4. Let A and B be **sets**, then the **set** $\mathcal{R} = \mathcal{P}(A \times B)$ of all **relations** between A and B and the **function space** $A \rightarrow \mathcal{P}(B)$ are **isomorphic**.

Proof. For each $R \in \mathcal{R}$ and each $x \in A$ we define: $S_x^R := \{y \in B \mid (x, y) \in R\}$ (the **set** of all y in B that are related to x by R). Example:

$$A = \{1, 2\} \quad B = \{a, b, c\} \quad R = \{(1, a), (1, c), (2, b)\}$$

Then for $x = 1 : S_1^R = \{a, c\}$ and for $x = 2 : S_2^R = \{b\}$.

We define a the following mapping:

$$\varphi : \mathcal{R} \rightarrow (A \rightarrow \mathcal{P}(B)), \quad \varphi(R) = f_R \quad \text{where } f_R(x) = S_x^R.$$

For our example, this would be $\varphi(R) = f_R = \{1 \mapsto \{a, c\}, 2 \mapsto \{b\}\}$

Then, for any $f \in A \rightarrow \mathcal{P}(B)$, we compute the **inverse** as $\varphi^{-1}(f) = \{(x, y) \mid y \in f(x)\}$. This is exactly the **relation** corresponding to f .

Since both directions match up perfectly, φ is a **bijection**, hence an **isomorphism**. □

Example: the **> relation** on $\mathbb{N}$ can also be represented as the **function** $f_{>}$ that maps a number n to the **set** $\{n+1, \dots\}$, i.e. $f_{>} : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N}) ; n \mapsto \{m \mid m > n\}$

Another example: let $A = \{x, y\}, B = \{1\}$, the **cartesian product** $A \times B = \{(x, 1), (y, 1)\}$. Hence, the **power set** (all possible relations) is:

$$\mathcal{P}(A \times B) = \{\{\phi\}, \{(x, 1)\}, \{(y, 1)\}, \{(x, 1), (y, 1)\}\}$$

Note: the **cardinality** of the **power set** of the **cartesian product** of two **sets** A, B is $\Gamma(\mathcal{P}(A \times B)) = 2^{\Gamma(A)\Gamma(B)}$.

$\mathcal{P}(B) = \{\{\phi\}, \{1\}\}$, **function space** $A \rightarrow \mathcal{P}(B)$ is the **set** of all **functions** $f : A \rightarrow \mathcal{P}(B)$:

$$A \rightarrow \mathcal{P}(B) = \{f_1, f_2, f_3, f_4\}$$

where:

- $f_1 = \{(x, \{\phi\}), (y, \{\phi\})\}$
- $f_2 = \{(x, \{\phi\}), (y, \{1\})\}$
- $f_3 = \{(x, \{1\}), (y, \{\phi\})\}$
- $f_4 = \{(x, \{1\}), (y, \{1\})\}$

Since $\Gamma(A \rightarrow \mathcal{P}(B)) = \Gamma(\mathcal{P}(A \times B))$, we can define a **bijection**:

- $\varphi(\{\phi\}) \mapsto f_1$
- $\varphi(\{(x, 1)\}) \mapsto f_3$
- $\varphi(\{(y, 1)\}) \mapsto f_2$

- $\varphi(\{(x, 1), (y, 1)\}) \mapsto f_4$

$\rightsquigarrow$ isomorphism.

Note

The bijection $\phi(R) = f$ entails the following properties of R :

- R is a **total relation**, iff $\phi \notin \text{Im}(f)$
- R is a **partial function**, iff $\Gamma(S) \leq 1 \quad \forall S \in \text{Im}(f)$
- $R = \phi$, iff $\text{Im}(f) = \{\phi\}$
- R is **reflexive**, iff $x \in f(x) \quad \forall x \in A$

We have defined **equivalence relations** as **reflexive**, **symmetric**, and **transitive** relations, e.g., **equality** is an **equivalence relation**.

Let S be a **set** of persons, and $=_A \subseteq S^2$, such that $s =_A t$, iff s and t have the same age, then $=_A$ is an **equivalence relation** on S .

Let S and T be **sets** and $S \sim_\Gamma T$, iff there is a **bijection** $f : S \rightarrow T$, then $\sim_\Gamma$ is an **equivalence relation**. For example, let $S = \{1, 2, 3\}$ $T = \{a, b, c\}$. We can define a **bijection** $f = \{1 \mapsto a, 2 \mapsto b, 3 \mapsto c\}$, so, $S \sim_\Gamma T$ (they have the same **size**).

Lemma B.5. If S is a **set** and $R \subseteq S^2$ is an **equivalence relation**, then $= \subseteq R$.

Proof Sketch: because of **reflexivity** ($\forall x \in S, (x, x) \in R$), the **equality relation** $=$ is exactly $\{(x, x) \mid x \in S\}$. Therefore, $= \subseteq R$.

B.4 Equivalence Classes

Def B.68. Equivalence Classes and Quotients: Let S be a [set](#) and R be an [equivalence relation on \$S\$](#) , then for any $x \in S$ we call the set $[x]_R := \{y \in S \mid R(x, y)\}$ the **equivalence class** of x under R , and the [set](#) $S/R := \{[x]_R \mid x \in S\}$ the **quotient space** of S under R , it is often read as S modulo R . The [element](#) x is called the **representative** of $[x]_R \in S/R$.

Example: Let $S = \mathbb{Z}$ and R is the [equivalence relation on \$S\$](#) that describes the same parity ([congruence modulo 2](#)) $x \equiv y \pmod{2} \Leftrightarrow 2 \mid (x - y)$:

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x \equiv y \pmod{2}\}$$

- $(x, y) \in R$ if the difference $x - y$ is divisible by 2.
- i.e. either both are even, or both are odd.

All odd-odd pairs:

$$\{(x, y) \mid x, y \in \{\dots, -3, -1, 1, 3, 5, \dots\}\}$$

All even-even pairs:

$$\{(x, y) \mid x, y \in \{\dots, -4, -2, 0, 2, 4, 6, \dots\}\}$$

Example pairs in R : $(1, 3), (5, -1), (7, 7), (2, 4), (0, -6), (-8, 10)$ Then, we have the following two distinct classes:

- $[1]_R = \{\dots, -3, -1, 1, 3, 5, \dots\}$
- $[2]_R = \{\dots, -4, -2, 0, 2, 4, 6, \dots\}$

For each $x \in \mathbb{Z}$ we have:

- the [set](#) $[1]_R$ as the [equivalence class](#) if x is odd
- the [set](#) $[2]_R$ as the [equivalence class](#) if x is even
- 1 and 2 are the [representatives](#)

And the [quotient space set](#) of $\mathbb{Z}$ under R is:

$$\mathbb{Z}/R = \{[1]_R, [2]_R\} = \{\text{odds, evens}\}$$

Def B.69. Canonical Projection: The mapping $\pi_R : S \rightarrow S/R$; $x \mapsto [x]_R$ is called the **canonical projection** or **canonical surjection** of S to S/R .

Given our example, we define the following [canonical projection](#):

$$\pi_R : \mathbb{Z} \rightarrow \{[1]_R, [2]_R\}, \quad x \mapsto [1]_R \text{ if odd, and } x \mapsto [2]_R \text{ if even.}$$

Def B.70. System of Representatives: Let R be an [equivalence relation on \$S\$](#) , a [subset](#) $M \subseteq S$ is called a system of representatives, iff M contains exactly one [representative](#) for each [equivalence class](#) of R .

Given our example, a [system of representatives](#) is any [subset](#) of $\mathbb{Z}$ that picks exactly one [element](#) from each [equivalence class](#), e.g.

- $M = \{1, 2\}$ is a [system of representatives](#)
- $M = \{-3, 0\}$ is a [system of representatives](#)
- $M = \{5, -8\}$ is a [system of representatives](#)

Observation: Often a [quotient space](#) S/R behaves similarly to the original [set](#) S .

Example: $n \equiv_k m \Leftrightarrow k \mid (n - m)$ is an [equivalence relation](#) and $\pi_{\equiv_k} : \mathbb{Z} / \equiv_k \rightarrow \{n \mid 0 \leq n \leq (k - 1)\}$ is [bijective](#).

Another example, the general [congruence modulo](#) $n \equiv_k m \Leftrightarrow k \mid (n - m)$ partitions $\mathbb{Z}$ into exactly k [equivalence classes](#) (all numbers that leave the same remainder when divided by k). e.g. $k = 3$:

- $[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$
- $[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$
- $[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}$

We have the **canonical projection** $\pi_{\equiv_k} : \mathbb{Z} \rightarrow \mathbb{Z}/\equiv_k, \quad n \mapsto [n]$ that sends each $n \in \mathbb{Z}$ to its **equivalence class**.

Notice that the **quotient space** $\mathbb{Z}/\equiv_k$ has exactly k classes, so, we have the following **bijection**:

$$\varphi : \mathbb{Z}/\equiv_k \rightarrow \{0, 1, \dots, k-1\}, \quad [n] \mapsto \text{the remainder of } n \text{ mod } k.$$

If we compose the **canonical projection** with the **bijection** we get the remainder, e.g. for $k = 3$:

$$-6 \xrightarrow{\pi_{\equiv_3}} [0] \xrightarrow{\varphi} 0.$$

$$7 \xrightarrow{\pi_{\equiv_3}} [1] \xrightarrow{\varphi} 1.$$

Lemma B.6. Let $=_{\subseteq} S \times S$ be the **equality relation** on a set S , then $[x]_{=} \in S/_{=}$ is always a **singleton** and thus $S \sim_{\Gamma} S/_{=}$.

Proof. If we take the **equality relation on a set** S , then for any $x \in S$, the **equivalence class** is:

$$[x]_{=} = \{y \in S \mid y = x\}$$

But the only y that satisfies $y = x$ is $y = x$ itself. So we get $[x]_{=} = \{x\}$. This is a **singleton set**. So every **equivalence class** under $=$ has exactly one element.

The **quotient space** is $S/_{=} := \{[x]_{=} \mid x \in S\}$ and each $[x]_{=}$ is the singleton set $\{x\}$, so the **quotient space** becomes $S/_{=} := \{\{x\} \mid x \in S\}$. In other words, the **quotient space** under the **equality relation** just replaces each **element** x with one-element **set** $\{x\}$.

That means the **set** S is in **bijection** with its **quotient space** under $=$ via the following **bijjective** map:

$$\phi : S \rightarrow S/_{=}, \quad x \mapsto [x]_{=} = \{x\}$$

Therefore, we express the **bijection** via same **cardinality**:

$$S \sim_{\Gamma} S/_{=}$$

□

Fun Example: Let S be the **set** of all cities worldwide and $\sim_c \in S^2$ defined by $a \sim_c b$, iff a and b are in the same country. Then $\sim_c$ is an **equivalence relation on** S , and the **quotient space** $S/\sim_c$ of S under $\sim_c$ consists of the **equivalence class** $[x]_{\sim_c}$ where $x \in S$. The capital of a country is a good **representative** for an **equivalence class**.

For instance: let's take a **subset** of all worldwide cities for simplicity, consider **domain of discourse**:

$$S = \{\text{Berlin, Munich, Hamburg, Paris, Lyon, Marseille, Rome, Milan}\}.$$

For $a, b \in S$ define the **relation**:

$$a \sim_c b \iff a \text{ and } b \text{ are in the same country.}$$

This **relation** is:

- **reflexive**: any city is in the same country as itself.
- **symmetric**: if Berlin is in the same country as Munich, then Munich is in the same country as Berlin.
- **transitive**: if Berlin and Munich are in the same country, and Munich and Hamburg are in the same country, then Berlin and Hamburg are in the same country.

Hence, $\sim_c$ is an **equivalence relation**.

The **quotient space** $S/\sim_c$ is the **set** of **equivalence classes**, i.e. the partition of S into countries:

$$[\text{Berlin}]_{\sim_c} = \{\text{Berlin, Munich, Hamburg}\}, \quad [\text{Paris}]_{\sim_c} = \{\text{Paris, Lyon, Marseille}\}, \quad [\text{Rome}]_{\sim_c} = \{\text{Rome, Milan}\}.$$

So

$$S/\sim_c = \left\{ \{\text{Berlin, Munich, Hamburg}\}, \{\text{Paris, Lyon, Marseille}\}, \{\text{Rome, Milan}\} \right\}.$$

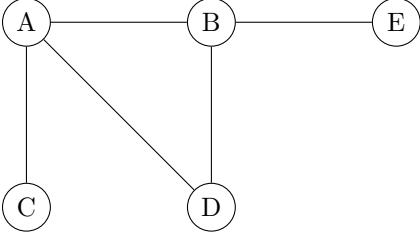
The **set** of chosen **representatives** is: $\{\text{Berlin, Paris, Rome}\}$.

C Graphs and Trees

Def C.1. Undirected Graph: An undirected graph is a pair $\langle V, E \rangle$ such that

- V is a **set** of **vertices (nodes)**, and
- $E \subseteq \{ \{v, \bar{v}\} \mid v, \bar{v} \in V \wedge v \neq \bar{v} \}$ is the **set** of its **undirected edges**.

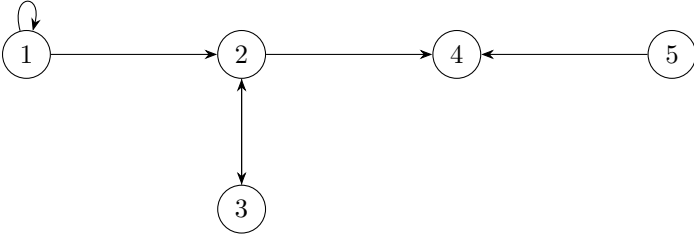
Example: An **undirected graph** $G_1 = \langle V_1, E_1 \rangle$, where $V_1 = \{A, B, C, D, E\}$ and $E_1 = \{\{A, B\}, \{A, C\}, \{A, D\}, \{B, D\}, \{B, E\}\}$



Def C.2. Directed Graph: A directed graph (digraph) is a pair $\langle V, E \rangle$ such that

- V is a **set** of **vertices (nodes)**, and
- $E \subseteq V \times V$ is the **set** of its **directed edges**.

Example: A **directed graph** $G_2 = \langle V_2, E_2 \rangle$, where $V_2 = \{1, 2, 3, 4, 5\}$ and $E_2 = \{(1, 1), (1, 2), (2, 3), (3, 2), (2, 4), (5, 4)\}$



Def C.3. Indegree: Given a **directed graph** $\langle V, E \rangle$, the indegree $\text{indeg}(v)$ of a vertex $v \in V$ is defined as $\text{indeg}(v) = \Gamma(\{w \mid (w, v) \in E\})$.

Example: $\text{indeg}(2) = 2$

Def C.4. Outdegree: Given a **directed graph** $\langle V, E \rangle$, the outdegree $\text{outdeg}(v)$ (branching factor) of a vertex $v \in V$ is defined as $\text{outdeg}(v) = \Gamma(\{w \mid (v, w) \in E\})$.

Example: $\text{outdeg}(5) = 1$

Note

For an **undirected graph**, $\text{indeg}(v) = \text{outdeg}(v)$ for all $v \in V$.

Observation: **directed graphs** are nothing else than **relations**.

Def C.5. Initial vs Terminal Node: Let $G = \langle V, E \rangle$ be a **directed graph**, then we call a node $v \in V$

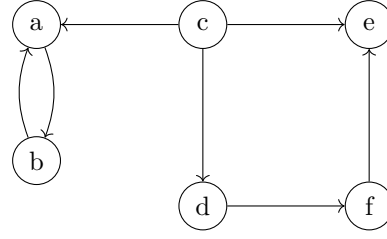
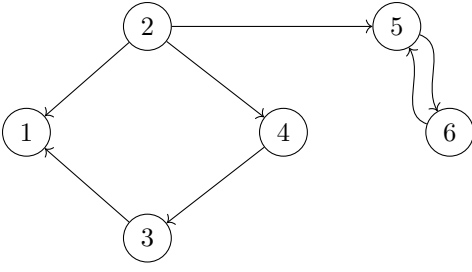
- **initial** (source of G), iff there is no $w \in V$ such that $(w, v) \in E$ (no predecessor)
- **terminal** (sink of G), iff there is no $w \in V$ such that $(v, w) \in E$ (no successor)

Def C.6. Graph Isomorphism: Iff we can find a **bijection** $\psi : V \rightarrow \bar{V}$ between two graphs $G = \langle V, E \rangle$ and $\bar{G} = \langle \bar{V}, \bar{E} \rangle$ then we call them isomorphic. The **bijection** $\psi : V \rightarrow \bar{V}$ is defined as $(a, b) \in E \Leftrightarrow (\psi(a), \psi(b)) \in \bar{E}$ for **directed graph**. And for **undirected graph** it is defined as $\{a, b\} \in E \Leftrightarrow \{\psi(a), \psi(b)\} \in \bar{E}$

Def C.7. Equivalent Graphs: Two graphs G and $\bar{G}$ are **equivalent** iff there is an **isomorphism** ψ between G and $\bar{G}$.

Example: the following two **graphs** are **equivalent** as there exists an **isomorphism** between them given by the **bijection**:

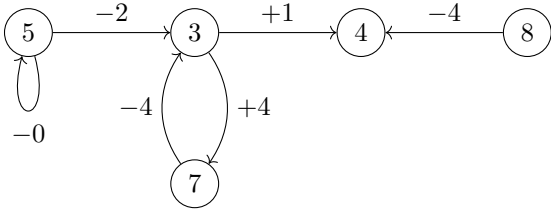
$$\psi := \{a \mapsto 5, b \mapsto 6, c \mapsto 2, d \mapsto 4, e \mapsto 1, f \mapsto 3\}$$



Def C.8. Labeled Graph: A labeled graph G is a quadruple $\langle V, E, L, l \rangle$ where $\langle V, E \rangle$ is a graph and $l : V \cup E \rightarrow L$ is a **partial function** into a **set** L of labels.

Example: $G = \langle V, E, L, l \rangle$ with $V = \{A, B, C, D, E\}$, where

- $E = \{(A, A), (A, B), (B, C), (C, B), (B, D), (E, D)\}$
- $l : V \cup E \rightarrow \{+, -, \phi\} \times \{1, \dots, 9\}$ with
 - $l(A) = 5 \quad l(B) = 3 \quad l(C) = 7 \quad l(D) = 4 \quad l(E) = 8,$
 - $l((A, A)) = -0 \quad l((A, B)) = -2 \quad l((B, C)) = +4,$
 - $l((C, B)) = -4 \quad l((B, D)) = +1 \quad l((E, D)) = -4$



Def C.9. Paths in Graphs: Given a graph $G := \langle V, E \rangle$ we call a $n + 1$ -tuple $p = \langle v_0, \dots, v_n \rangle \in V^{n+1}$ a **path** in G iff $(v_{i-1}, v_i) \in E$ for all $1 \leq i \leq n$ and $n > 0$.

- We say that v_i are nodes on p and that v_0 and v_n are **linked** by p .
- v_0 and v_n are called the **start** and **end** of p (write $\text{start}(p)$ and $\text{end}(p)$), the other v_i are called **inner nodes** of p .
- n is called the **length** of p (write $\text{len}(p)$).
- We denote the **set** of paths in G with $\Pi(G)$.

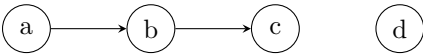
Note

Not all v_i -s in a **path** are necessarily different.

For a graph $G = \langle V, E \rangle$ and a **path** $p = \langle v_1, \dots, v_n \rangle \in V^{n+1}$, write

- $v \in p$, iff $v \in V$ is a vertex on the **path** ($\exists i. v_i = v$)
- $e \in p$, iff $e = (v, \bar{v}) \in E$ is an edge on the **path** ($\exists i. v_i = v \wedge v_{i+1} = \bar{v}$)

Example: Consider $G = \langle V, E \rangle$ with $V = \{a, b, c, d\}$ and $E = \{(a, b), (b, c)\}$:



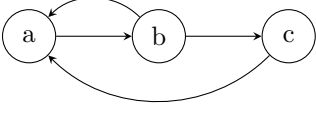
An example **path** would be $p = \langle a, b, c \rangle$. Here $n = 2$ and $p \in V^3 = \{(a, a, a), (a, b, d), \dots\}$. Obviously, since $n = 2$ so we have 2 edges and 3 nodes.

- nodes a, b, c are on the **path**
- $\text{start}(p) = a$ and $\text{end}(p) = c$
- b is an inner node, and $\text{len}(p) = 2$
- $e = (a, b), \bar{e} = (b, c) \in p$
- $\Pi(G) = \{(a, b), (b, c), (a, b, c)\}$

Def C.10. Cyclic Graphs: Given a **directed graph** $G = \langle V, E \rangle$, a **path** p is called **cyclic** iff $\text{start}(p) = \text{end}(p)$. A cycle $\langle v_0, \dots, v_n \rangle$ is called **simple**, iff $v_i \neq v_j$ for $1 \leq i, j \leq n$ with $i \neq j$ (all inner nodes are distinct).

Def C.11. DAG: A **directed graph** with no **cycles** is called **directed acyclic graph (DAG)**.

Example: $p = \langle a, b, c, a \rangle$ is simple **cyclic** path. $p = \langle a, b, a, b, a \rangle$ is a **cyclic** path that is not simple.



Def C.12. Node Depth: Let $G = \langle V, E \rangle$ be a **directed graph**, then the depth $\text{dp}(v)$ of a vertex $v \in V$ is defined to be 0, iff v is a source of G and the **supremum** $\sup(\{\text{len}(p) \mid \text{indeg}(\text{start}(p)) = 0 \wedge \text{end}(p) = v\})$ otherwise, i.e. the length of the longest **path** from a source of G to v (can be infinite).

Def C.13. Graph Depth: Given a **directed graph** $G = \langle V, E \rangle$, the **depth** ($\text{dp}(G)$) of G is defined as the **supremum** $\sup(\{\text{len}(p) \mid p \in \Pi(G)\})$, i.e. the maximal path length in G .

Def C.14. Tree: A tree is a **DAG** $G = \langle V, E \rangle$ such that

- There is exactly one **initial node** $v_r \in V$ (called the **root**)
- All nodes but the root have **indegree** of 1.

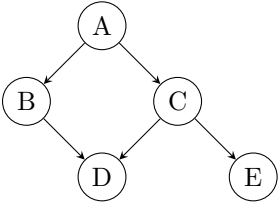
We call v the **parent** of w , iff $(v, w) \in E$ (w is a child of v). We call a node v a **leaf** of G , iff it is **terminal**, i.e. if it does not have children.

Lemma C.1. For any node $v \in V$ except the root v_r , there is exactly one **path** $p \in \Pi(G)$ with $\text{start}(p) = v_r$ and $\text{end}(p) = v$.

Def C.15. Non-Descendants: Let $\langle X, E \rangle$ be a **DAG**, $x \in X$, and E^* the **reflexive transitive closure** of E . The **non-descendants** of x are the **elements** of the **set** $\text{NonDesc}(x) := \{y \mid (x, y) \notin E^*\} \setminus \text{Parents}(x)$

Example: Consider the following **DAG** $G = \langle X, E \rangle$ where:

- $X = \{A, B, C, D, E\}$
- $E = \{(A, B), (A, C), (B, D), (C, D), (C, E)\}$
- $E^* = \{(A, A), (B, B), (C, C), (D, D), (E, E), (A, B), (A, C), (A, D), (A, E), (B, D), (C, D), (C, E)\}$



Let's find the **non-descendants** of node B :

- First, we find all y such that $(B, y) \in E^*$. Looking at E^* , the only pairs starting with B are (B, B) and (B, D) . Thus, the reachable nodes from B are $\{B, D\}$.
- The set of nodes y where $(B, y) \notin E^*$ is everything else: $\{A, C, E\}$.
- The parents of B is the set $\text{Parents}(B) = \{A\}$.
- By the definition, we exclude the parents: $\text{NonDesc}(B) = \{A, C, E\} \setminus \{A\} = \{C, E\}$.

D Mathematical Structures

Def D.1. Mathematical Structure: A mathematical structure combines multiple [mathematical objects](#) (the components) into a new [object](#). The components usually have names by which they can be referenced. Given a [definition](#) of a mathematical structure S , we say that any [object](#) that conforms to that is an instance of S .

Example: A graph $G = \langle V, E \rangle$ is a [mathematical structure](#) where the components are: V (the [set](#) of vertices) and E (the [set](#) of edges). A graph G where $V = \{1, 2, 3\}$ and $E = \{(1, 2), (2, 3)\}$ is one instance of a graph.

Def D.2. Structure Signature: A structure signature combines the components, their types, and accessor names of a [mathematical structure](#) in a tabular overview.

Example:

$$\left\langle \begin{array}{ll} V & \text{Set} \\ E & \{(u, v) \mid u, v \in V\} \end{array} \begin{array}{l} \text{vertices (nodes)} \\ \text{edges} \end{array} \right\rangle$$

Note

Most programming languages have some way of creating "*named structures*" and referencing components is usually done via "*dot notation*" (e.g. `person.name`).

Def D.3. Group: A group is a [mathematical structure](#) $\langle G, \circ, e, .^{-1} \rangle$ that consists of:

- a base [set](#) G of [objects](#),
- an operation $\circ : G \times G \rightarrow G$, such that $\forall a, b \in G. a \circ b \in G$ and $\forall a, b, c \in G. (a \circ b) \circ c = a \circ (b \circ c)$,
- a unit $e \in G$, such that $\forall a \in G. e \circ a = a \circ e = a$, and
- the [inverse function](#) $.^{-1} : G \rightarrow G$, such that $\forall a \in G. a \circ a^{-1} = a^{-1} \circ a = e$

An example of a [group](#) is the [set](#) $G = \mathbb{Z}$, with the operation $+: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$; then $e = 0$ and $.^{-1}$ is $\lambda x \in \mathbb{Z}. -x$. We write it as $\langle \mathbb{Z}, +, 0, - \rangle$.

Question: can we do the same for multiplication where $e = 1$? If we think about it, we would be stuck on finding the $.^{-1}$ component of the group. What can we always multiply with any integer and get $e = 1$ back? Take for example 1, we can multiply it with $\frac{1}{1}$ (great!). Take -1 , we can multiply it with $\frac{1}{-1}$ (also great!). But take any other integer, e.g. 2, we must multiply it with $\frac{1}{2}$ to get 1 back, however, $\frac{1}{2} \notin \mathbb{Z}$, it is however $\in \mathbb{Q}$! Moreover, 0 never has a multiplicative inverse in any field! The multiplicative group would only make sense iff $G = \mathbb{Q} \setminus \{0\}$. We write that as $\langle \mathbb{Q} \setminus \{0\}, *, 1, a \mapsto \frac{1}{a} \rangle$.

But we need some [multiplication structure](#) that works for $\mathbb{Z}$!

Def D.4. Monoid: A monoid is a [mathematical structure](#) $\langle M, \circ, e \rangle$ that consists of:

- A base set M of [objects](#),
- An operation $\circ : M \times M \rightarrow M$, such that for all $a, b \in M$, we have $a \circ b \in M$ (closure),
- Associativity: for all $a, b, c \in M$, $(a \circ b) \circ c = a \circ (b \circ c)$, and
- A unit element $e \in M$, such that for all $a \in M. e \circ a = a \circ e = a$.

Hence, the [mathematical structure](#) $\langle \mathbb{Z}, *, 1 \rangle$ is a [monoid](#) (a [group](#) missing an [inverse function](#)).

We saw that $\langle \mathbb{Z}, +, 0, - \rangle$ is a [group](#) under addition. And $\langle \mathbb{Z}, *, 1 \rangle$ is a [monoid](#) under multiplication. Now, we want to do everyday arithmetics with integers; we want expressions where we have additions and [multiplication](#) (e.g. $a * (b + c)$) which evaluates to $(a * b) + (a * c)$.

For that, we need a [mathematical structure](#) where both addition and [multiplication](#) live together.

Def D.5. Ring: A ring is a [mathematical structure](#) $\langle R, +, 0, -, *, 1 \rangle$ consisting of:

- A base [set](#) R .
- An addition operation $+: R \times R \rightarrow R$ such that $\langle R, +, 0, - \rangle$ is a [group](#) (called abelian [group](#)).
- A multiplication operation $*: R \times R \rightarrow R$ such that $\langle R, *, 1 \rangle$ is a [monoid](#).
- Distributivity: for all $a, b, c \in R, a * (b + c) = (a * b) + (a * c), (a + b) * c = (a * c) + (b * c)$

Hence, mixing components from the **group** $\langle \mathbb{Z}, +, 0, - \rangle$ and the **monoid** $\langle \mathbb{Z}, *, 1 \rangle$ leaves us with a **ring** $\langle \mathbb{Z}, +, 0, -, *, 1 \rangle$

Def D.6. Magma: A magma is a **mathematical structure** $\langle M, \circ \rangle$ consisting of:

- A base **set** M , and
- A binary operation $\circ : M \times M \rightarrow M$.

Example: $\langle \mathbb{Z}, - \rangle$ (integers under subtraction). $\langle \mathbb{N}, + \rangle$ (natural numbers under addition)

Def D.7. Semigroup: A semigroup (**magma** with associativity) is a **mathematical structure** $\langle S, \circ \rangle$ consisting of:

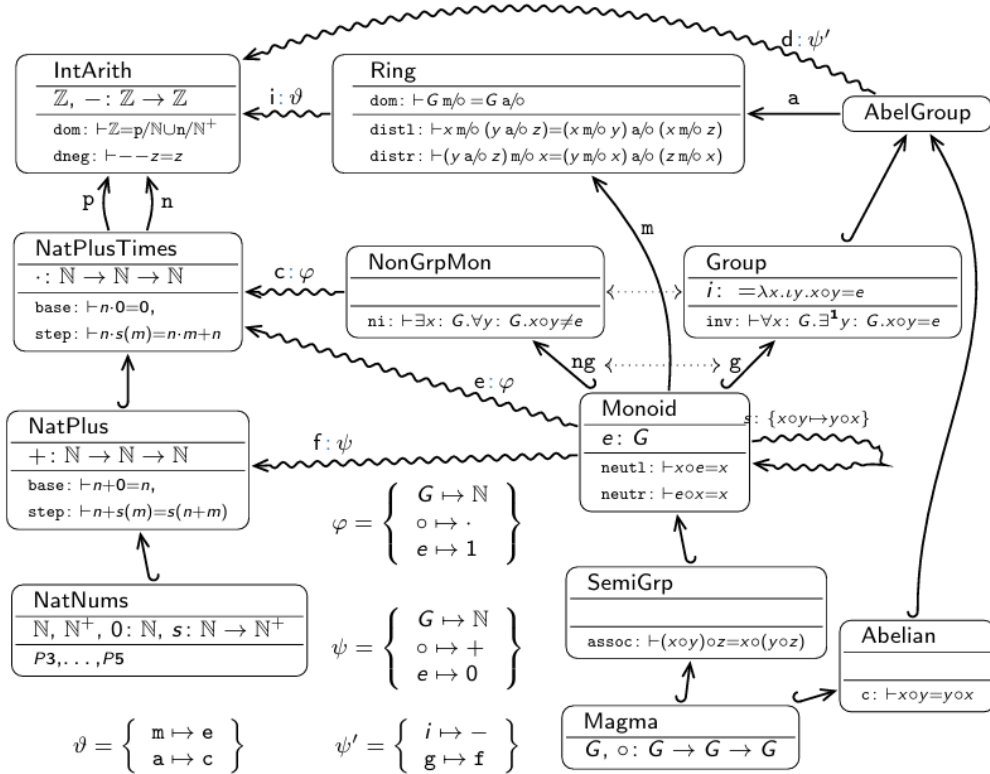
- A base **set** S ,
- A binary operation $\circ : S \times S \rightarrow S$, and
- Associativity: for all $a, b, c \in S$, $(a \circ b) \circ c = a \circ (b \circ c)$

Example: $\langle \mathbb{Z}, + \rangle$ (integers under addition).

Note

Given collections M and S of instances of **magmas** and **semigroups**, $S \subseteq M$ as there are **magmas** whose operation is not associative.

Mathematical Structures in a Graph:



- nodes are **structures** with **objects** ($o: \tau$) and properties ($n \vdash P$).
- edges represent inheritance
 - regular inheritance: $\hookrightarrow$
 - copying substructures: $\xrightarrow{n}$
 - interpretation via φ : $\xrightarrow{\varphi}$
- Examples are just interpretations ($\rightsquigarrow$) read backwards
- Any **object** / **property** / **theorem** can be copied along any edge

E Formal Languages and Grammars

Def E.1. Alphabet: An alphabet is a [finite set](#); we call each element $a \in A$ a **character**, and an n -tuple $s \in A^n$ a **string** (of length n over A).

Note

We often write a string $\langle c_1, \dots, c_n \rangle$ as " $c_1 \dots c_n$ ", for instance "abc" for $\langle a, b, c \rangle$

Example: Let $A = \{0, 1\}$ be an [alphabet](#)

- "0", "1" are strings of length 1 ($A^1 = \{0, 1\}$)
- "00", "01", "10", "11" are strings of length 2 in (A^2)
- "001" $\in A^3$

Def E.2. Empty String: Let A be an [alphabet](#), then $A^0 = \{\langle \rangle\}$, where $\langle \rangle$ is the unique 0-tuple. We consider $\langle \rangle$ as the string of length 0 and call it the **empty string** and denote it with ϵ .

Note

$\{1, 2, 3\} = \{3, 2, 1\}$ BUT $\langle 1, 2, 3 \rangle \neq \langle 3, 2, 1 \rangle$ ([sets](#) vs strings)

Def E.3. String Length: Given a string s , we denote its length with $|s|$.

Def E.4. String Concatenation: The concatenation $\text{conc}(s, t)$ of two strings $s = \langle s_1, \dots, s_n \rangle \in A^n$ and $t = \langle t_1, \dots, t_m \rangle \in A^m$ is defined as $s + t$ or simply st

Example: $\text{conc}(\text{"text"}, \text{"book"}) = \text{"text"} + \text{"book"} = \text{"textbook"}$.

Def E.5. Kleene Plus and Kleene Star: Let A be an [alphabet](#), then we define the [sets](#):

- $A^+ := \bigcup_{i \in \mathbb{N}^+} A^i$ (*nonempty strings*), and
- $A^* := A^+ \cup \{\epsilon\}$ (*strings*).

Def E.6. Formal Language: A [set](#) $L \subseteq A^*$ is called a *formal language* over A .

Example: If $A = \{a, b\}$ then:

$$A^* = \bigcup_{n=0}^{\infty} A^n$$

where $A^0 = \epsilon$, $A^1 = \{a, b\}$, $A^2 = \{a, b, ab, ba\}$, and so on ...

$$A^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, \dots\}$$

$$A^+ = \{a, b, aa, ab, ba, bb, aaa, aab, aba, abb, \dots\}$$

A [formal language](#) might be $L = \{a, b, aab, bbb, \dots\}$

Def E.7. Repeating Chars: We use $c^{[n]}$ for the string that consists of the character c repeated n times.

Examples:

- Let $A = \{a, b\}$, $a^{[5]} = \text{"aaaaa"}$.
- The [set](#) $M := \{ba^{[n]} \mid n \in \mathbb{N}\}$ of strings that start with character b followed by an arbitrary numbers of a 's is a [formal language](#) over $A = \{a, b\}$

Note

A [formal language](#) might be infinite and even undecidable even if the [alphabet](#) A is [finite](#). For example, let $A = \{a, b\}$ then the [formal language](#) $L = \{a, ab, bba\}$ is [finite](#) but the [formal language](#) $L = \{a^{[n]} \mid n \in \mathbb{N}\}$ is infinite.

Since we cannot list all strings in a **formal language** L because there are infinitely many, we need a **finite description** that can generate all strings in L . We need **Grammars**.

Def E.8. Phrase Structure Grammar: A phrase structure grammar (*type 0 grammar, unrestricted grammar, grammar*) is a tuple $\langle N, \Sigma, P, S \rangle$ where

- N is a **finite set** of **nonterminal symbols**,
- Σ is a **finite set** of **terminal symbols**. (members of $N \cup \Sigma$ are called symbols).
- P is a **finite set** of **production rules**: pairs $p := h \rightarrow b$ (also written as $h \Rightarrow b$), where $h \in (\Sigma \cup N)^* N (\Sigma \cup N)^*$ and $b \in (\Sigma \cup N)^*$. The string h is called the **head** of p and b the **body**.
- $S \in N$ is a distinguished symbol called the **start symbol** (also **sentence symbol**).

The **sets** N and Σ are assumed to be **disjoint**. Any word $w \in \Sigma^*$ is called a **terminal word**.

Note

Production rules map strings with at least one **nonterminal** to arbitrary other strings

Notation: If we have n **production rule** $h \rightarrow b_i$ sharing a head, we often write $h \rightarrow b_1 \mid \dots \mid b_n$ instead.

Example: A simple **grammar** for english sentences:

$$\begin{aligned} S &::= NP \text{ Vi} \\ NP &::= \text{Article } N \\ \text{Article} &::= \text{the} \mid \text{a} \mid \text{an} \\ N &::= \text{dog} \mid \text{teacher} \mid \dots \\ \text{Vi} &::= \text{sleeps} \mid \text{smells} \mid \dots \end{aligned}$$

Here S is the **start symbol**, NP , Article , N , and Vi are **nonterminals**, the , a , dog , smells , ... are **terminals**. Let's generate a valid sentence from the **grammar**:

NP	Vi
$\text{Article } N$	Vi
$\text{the } N$	Vi
$\text{the } \text{dog}$	Vi
$\text{the } \text{dog}$	sleeps

Def E.9. Lexical: A **production rule** whose head is a single **nonterminal** and whose body consists of a single **terminal** is called **lexical** or a **lexical insertion rule**.

Example: Both of the following **rules** are **lexical**

$$\begin{aligned} \text{Noun} &\rightarrow \text{"dog"} \\ \text{Verb} &\rightarrow \text{"run"} \end{aligned}$$

Def E.10. Lexicon of a Grammar: The **subset** of **lexical rules** of a **grammar** G is called the **lexicon** of G

Def E.11. Vocabulary: The **set** of body symbols are called the **vocabulary** (or the **alphabet**).

Def E.12. Lexical Categories of a Grammar: The **nonterminals** that appear in the heads of **lexical** rules are called **lexical categories** of the **grammar**.

Def E.13. Structural Rules: The non **lexicon production rules** are called **structural**.

Def E.14. Phrasal categories: The **nonterminals** in the heads of **production rules** that expand into other **nonterminals** are called **phrasal** (syntactic) categories.

Def E.15. G -Derivation: Given a **phrase structure grammar** $G := \{N, \Sigma, P, S\}$, we say G **derives** $t \in (\Sigma \cup N)^*$ from $s \in (\Sigma \cup N)^*$ in one step, iff there is a **production rule** $p \in P$ with $p = h \rightarrow b$ and there are $u, v \in (\Sigma \cup N)^*$, such that $s = uhv$ and $t = ubv$. We write $s \rightarrow_G^p t$ (or $s \rightarrow_G t$ if p is clear from the context) and use $\rightarrow_G^*$ for the **reflexive transitive** closure of $\rightarrow_G$. We call $s \rightarrow_G^* t$ a G -derivation of t from s .

Let's unravel that. Take the following **grammar**:

- $N = \{S\}$

- $\Sigma = \{a, b\}$
- $P = \{S \rightarrow aSb, S \rightarrow ab\}$
- $S = S$

The following derivations holds:

- $S \xrightarrow{p}_G aSb$ where $p : S \rightarrow aSb$ (*one step*)
- $aSb \xrightarrow{q}_G aabb$ where $q : S \rightarrow ab$ (*one step*)
- $S \xrightarrow{S \rightarrow aSb}_G aSb \xrightarrow{S \rightarrow ab}_G aabb$ (*multiple steps*). So, $S \xrightarrow{*}_G aabb$

Def E.16. Sentential Form: Given a **phrase structure grammar** $G := \{N, \Sigma, P, S\}$, we say that $s \in (\Sigma \cup N)^*$ is a **sentential form** of G , iff $S \xrightarrow{*}_G s$

Def E.17. Sentence: A **sentential form** that does not contain **nonterminals** is called a sentence of G , we also say that G **accepts** s . We say that G **rejects** s , iff it is not a sentence of G .

Examples

1. "Article teacher Vi" is a **sentential form**

$$\begin{aligned} S &\xrightarrow{p}_G \text{NP Vi} \\ &\xrightarrow{q}_G \text{Article N Vi} \\ &\xrightarrow{q}_G \text{Article teacher Vi} \end{aligned}$$

2. "the teacher sleeps" is a **sentence**

$$\begin{aligned} S &\xrightarrow{*}_G \text{Article teacher Vi} \\ &\xrightarrow{q}_G \text{the teacher Vi} \\ &\xrightarrow{q}_G \text{the teacher sleeps} \end{aligned}$$

Def E.18. Language Generation: The language $L(G)$ of G is the **set** of its **sentences**. We say that $L(G)$ is **generated** by G .

Def E.19. Equivalent Grammars: We call two **grammars** **equivalent**, iff they have the same **languages**.

Def E.20. Universal Grammar: A **grammar** G is said to be **universal** if $L(G) = \Sigma^*$

Def E.21. Syntactic Analysis: Syntactic analysis (parsing / syntax analysis) is the process of analyzing a string of symbols, either in a **formal** or a **natural language** by means of a **grammar**.

Note

The shape of the **grammar** determines the size of its **language**

Def E.22. Context-Sensitive: We call a **grammar** context-sensitive (or type 1), if the bodies of **production rules** have no less symbols than the heads.

Example: The **language** $\{a^{[n]}b^{[n]}c^{[n]}\}$ is **accepted** by

$$\begin{aligned} S &\rightarrow a b c \mid A \\ A &\rightarrow a A B c \mid a b c \\ c B &\rightarrow B c \\ b B &\rightarrow b b \end{aligned}$$

Def E.23. Context-Free: We call a **grammar** context-free (or type 2), iff the heads have exactly one symbol.

Example: The **language** $\{a^{[n]}b^{[n]}\}$ is **accepted** by $S \rightarrow a S b \mid \epsilon$

Def E.24. Regular: We call a **grammar** regular (or type 3, or REG), if additionally the bodies are empty or consist of a **nonterminal**, optionally followed by a **terminal symbol**.

Example: The **language** $\{a^{[n]}\}$ is **accepted** by $S \rightarrow S a \mid \epsilon$

Note

By extension, a [formal language](#) L is called *context-sensitive* / *context-free* / *regular*, iff it is the [language](#) of a respective [grammar](#). [Context-free grammars](#) are sometimes **CFGs** and context-free languages **CFLs**.

Observation: [Natural languages](#) are probably [context sensitive](#) but parsable in real time!